

Information System Analysis & Design on Chaldal Company



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Chapter 1

Recognition of Need

1.1 Introduction

A system is a set of interconnected elements, components, or individuals organized to achieve a specific goal. Systems design focuses on enhancing performance by identifying necessary improvements, forming architectures, modules, and data to meet specific needs. This approach implements systems theory in product design. System analysis involves gathering information about an organization, defining problems, and suggesting potential solutions. The goal is to improve the system's functionality by addressing internal issues and recommending actionable changes. In this context, Chaldal Limited, an online grocery shop, is analyzed to identify its system's internal problems and propose recommendations for improvement.

Chaldal.com, founded in 2013, is a grocery e-commerce platform in Bangladesh. The company offers its services via a website and mobile apps for Android and iOS. "Chaldal" is a Bengali word colloquially used to refer to "groceries" (Chal means Rice, and Dal means Pulses). Chaldal started as an online grocery service provider and has since gone on to redefine supply chains, ease commodity trade, support refugee camps, and reduce food wastage by building technology into the supply chain, all the way back to the farms. Chaldal operates in Dhaka, Narayanganj, Chittagong, Jashore, Khulna, Sylhet, and Rajshahi. As of February 2022, it has a total of 28 active warehouses. Chaldal operates using two distinct E-business models:

1. Business to Consumer (B2C) Model:

In the B2C model, Chaldal sells its products directly to consumers. Customers can browse the products displayed on Chaldal's website, select the items they want, and place an order. Once the order is received, Chaldal is notified via email and promptly dispatches the goods to the customer.

2. Business to Business (B2B) Model:

Chaldal also employs the B2B model, where it sells products to an intermediary buyer, who then resells the items to the final consumer. For instance, a wholesaler might order products from Chaldal's website, and upon receiving the shipment, they sell the products to the end customers who visit the wholesaler's retail outlet.

Chaldal, established by Waseem Alim (CEO), Zia Ashraf (COO), and Tejas Viswanath (CTO), entered the market during the early stages of e-commerce in Bangladesh when online grocery shopping was still a novel concept in Dhaka. The company operates with a compact yet efficient engineering team of 30 engineers who work across various domains, including Customer Experience, Logistics, Operations, Infrastructure, and Platform. These team members are distributed globally, with representatives in Bangladesh, the US, Singapore, New Zealand, Japan, Switzerland, and Australia. Overall, Chaldal has grown into a diverse team of over 1,000 members, including delivery personnel. The company's current management structure can be categorized into three distinct levels: Administrative, Executive, and Operational.

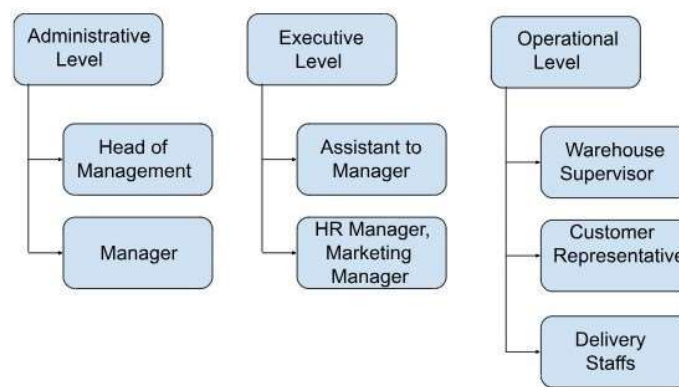


Figure 1.1: Employee Hierarchy of Chaldal.com

1.2 Problem Identification

Identifying problems within a system is a crucial step in system analysis, as it helps pinpoint areas that need improvement and guides the development of effective solutions. For Chaldal.com, several key problems have been identified across different operational aspects that directly impact user experience, efficiency, and overall business performance. Below are the defined problems:

Problem 01: Inefficient User Interface and Navigation

Chaldal.com's user interface and navigation are not optimized, making it challenging for users to locate products and complete purchases easily. The website's complex layout, coupled with slow loading times, leads to a frustrating user experience that may drive potential customers away.

Problem 02: Inconsistent Return and Refund Policies

The current return and refund policies of Chaldal.com are inconsistent and lack transparency, resulting in customer confusion and dissatisfaction. Unclear procedures and delayed processing of refunds not only erode customer trust but also impact the platform's reputation negatively.

Problem 03: Security Vulnerabilities

Chaldal.com faces significant security threats, including vulnerabilities to malware and unauthorized access. The platform's insufficient digital security measures put customer data at risk, compromising the integrity of the system and threatening the overall business operation.

Problem 04: Dynamic Pricing Challenges

Chaldal.com struggles with dynamic pricing issues, where the frequent fluctuation of prices causes confusion and mistrust among customers. The inconsistency in pricing across platforms and a lack of clear communication about price changes create a negative perception, affecting customer loyalty.

Problem 05: Order Delivery Disruptions

Order delivery disruptions pose a significant obstacle for Chaldal.com, resulting in not only frequent delays, incorrect deliveries, and inadequate communication but also escalating operational costs and inefficiencies. Addressing these challenges is crucial to

enhancing customer experience, fostering loyalty, and maintaining a competitive edge in the market.

1.3 Conclusion

This chapter highlights the essential problems facing Chaldal.com, underscoring the critical need for improvements across various operational and service areas. Through detailed identification of user experience challenges, inconsistencies in return and refund policies, digital security vulnerabilities, dynamic pricing complexities, and order delivery disruptions, it is evident that these issues significantly impact customer satisfaction, trust, and overall business performance.

Recognizing these needs is the first and most crucial step toward developing a robust and efficient system that can meet customer expectations and enhance Chaldal.com's market position. Addressing these problems with targeted solutions will not only improve operational efficiency but also help build a more reliable and user-friendly platform. Ultimately, acknowledging and responding to these identified needs will drive Chaldal.com towards achieving sustained growth, enhanced customer loyalty, and a competitive edge in the e-commerce landscape.

Chapter 2

Initial Feasibility Study

2.1 Introduction

The recognition of need is the initial step in system analysis, focusing on identifying the core problems or opportunities that require a solution. This stage establishes the rationale for developing or enhancing a system and involves close collaboration between the system analyst and various stakeholders, such as users, managers, and IT staff. Through interviews, surveys, and observations, the analyst gathers essential information about the system's requirements and objectives. This phase involves identifying specific problems, exploring opportunities to enhance operations, and understanding stakeholder needs. Requirement gathering is crucial, encompassing both functional and non-functional aspects, such as performance standards and security needs. Additionally, a feasibility assessment evaluates technical, economic, operational, and legal factors to ensure the project's viability, setting the foundation for the system's development.

2.2 Initial Feasibility Analysis

The Initial Feasibility Analysis is a crucial early step in the system analysis process, aimed at assessing the practicality and viability of the proposed system or project. This phase helps determine whether the project should proceed by evaluating various aspects that could impact its success. It involves a preliminary review of several factors to ensure that

potential challenges are identified and addressed before significant resources are invested.

The Initial Feasibility Analysis typically encompasses the following evaluations:

1. **Technical Feasibility:** Determines if the necessary technology and resources are available or can be developed. It assesses technical requirements such as hardware, software, and expertise, and identifies any potential technological constraints.
2. **Economic Feasibility:** Also known as cost-benefit analysis, this evaluates the financial aspects of the project. It compares the projected costs of development and maintenance with the expected benefits to ensure the investment is financially sound.
3. **Operational Feasibility:** Assesses if the system aligns with the organization's operational capabilities and goals. It examines how well the system integrates with current workflows and whether it will be supported effectively by users and staff.
4. **Legal Feasibility:** Reviews legal and regulatory issues that could impact the project, including compliance with data protection laws and industry regulations, to ensure the system meets all legal requirements.
5. **Schedule Feasibility:** Checks if the proposed timeline for the project is realistic and achievable, considering deadlines, resource availability, and potential risks that could cause delays.

2.2.1 Problem 01

2.2.1.1 Statement of the problem

Chaldal.com experiences user experience issues that affect customer satisfaction. Problems such as slow website loading times and complex navigation can lead to frustration, decreased user engagement, and potentially lost sales.

2.2.1.2 Summary of findings

The primary issue identified is that slow page load times and complicated site navigation negatively impact the overall user experience. This can lead to users abandoning

their shopping carts or leaving the site entirely, resulting in lost revenue and customer dissatisfaction. Additionally, a frustrating browsing experience can deter potential customers from returning to the site in the future, further decreasing sales opportunities. Addressing these issues is crucial for improving customer engagement and loyalty in a competitive e-commerce landscape.

2.2.1.3 Details of Findings

Chaldal.com's user experience is significantly compromised by two primary factors: slow page load times and a complex navigation structure. These issues are particularly pronounced during peak traffic periods, when users experience frustrating delays that can severely impact their overall satisfaction. When attempting to access product pages or complete transactions, many users encounter latency that leads to increased abandonment rates. This is concerning, as it not only disrupts the shopping experience but also discourages repeat visits.

The slow loading times create a sense of impatience among users, leading them to abandon their carts in favor of faster, more efficient competitors. Research shows that even a one-second delay in loading time can result in a significant drop in conversions, highlighting the critical need for performance enhancements. This challenge is exacerbated when users are trying to complete time-sensitive tasks, such as purchasing items during sales or promotions, further increasing the likelihood of cart abandonment.

In addition to loading delays, the site's navigation system is cumbersome and lacks intuitiveness. Users often find it difficult to locate specific products due to poorly organized categories and ineffective search functionalities. The navigation menus can be overwhelming, with too many options presented in a disorganized manner. This complexity leads to confusion and frustration, causing users to spend more time searching for products than actually enjoying the shopping experience.

The search function, a crucial component of any e-commerce platform, is not optimized for user needs. Users may enter search queries expecting to find relevant results, only to be met with a list of unrelated items or a complete lack of results. This inefficiency not only aggravates users but also diminishes their trust in the site's ability to meet their needs. So, addressing the issues of slow page load times and complex navigation is crucial

for enhancing the user experience on Chaldal.com

2.2.1.4 Recommendations

To address these user experience problems, the following recommendations are proposed:

Optimize Loading Times: Implement performance enhancements to reduce page load times, such as optimizing images, leveraging browser caching, and reducing server response times.

Improve Navigation: Revise the site's navigation structure to make it more intuitive and user-friendly. This could involve restructuring categories, improving the search functionality, and simplifying the user interface.

Enhance Usability Testing: Conduct regular usability testing to identify and address any issues that users may encounter. Gathering feedback from users can provide valuable insights into areas for improvement.

Monitor and Analyze Performance: Continuously monitor site performance and user behavior using analytics tools. This will help in identifying performance bottlenecks and areas where users may struggle, allowing for ongoing optimization.

2.2.2 Problem 02

2.2.2.1 Statement of the Problem

Chaldal.com faces challenges with its return and refund policy, leading to customer dissatisfaction and mistrust. Issues such as unclear return procedures and delays in processing refunds contribute to a negative customer experience and potential loss of repeat business.

2.2.2.2 Summary of Findings

The return and refund process on Chaldal.com is problematic, with customers reporting confusion over policy details and delays in receiving refunds. This lack of clarity and inefficiency in handling returns negatively impacts customer satisfaction and trust.

2.2.2.3 Details of Findings

Chaldal.com's return and refund policy has emerged as a significant challenge affecting customer satisfaction and overall trust in the brand. Many customers perceive the policy details as ambiguous and difficult to navigate, leading to confusion when they need to return products or request refunds. This lack of clarity not only frustrates customers but also creates a barrier to completing their purchases confidently.

Customers frequently report feelings of frustration due to the absence of clear guidelines on how to initiate returns and the specific steps involved in the refund process. For instance, unclear instructions on what items are eligible for return, the timeframe for initiating returns, and how to package items for return can all contribute to a negative experience. Additionally, customers often encounter delays in processing refunds, which prolongs their waiting period and exacerbates their dissatisfaction.

The inefficiencies in handling returns and refunds reflect poorly on Chaldal.com's customer service standards. In today's competitive e-commerce landscape, customers expect a straightforward and timely process for returns and refunds. When these expectations are not met, it can significantly detract from their overall shopping experience. The current system, characterized by unclear communication and processing delays, falls short of these essential customer service benchmarks.

Furthermore, the lack of effective communication regarding return procedures can lead to misunderstandings. For example, if customers are not informed about the specific documentation required for returns or if there are inconsistent policies regarding different product categories, it can create additional frustration. This ambiguity not only affects immediate customer satisfaction but also diminishes their trust in the company. When customers feel uncertain about how to resolve issues, they are less likely to engage with the brand in the future. To effectively tackle these challenges, Chaldal.com must prioritize the streamlining of its return and refund process.

2.2.2.4 Recommendation

Clarify Return Policy: Make the return policy clear and detailed on the website, outlining procedures, eligibility, and timelines to prevent customer confusion.

Streamline Refund Process: Improve the refund process to minimize delays by automating parts of the system and expediting transaction handling.

Enhance Communication: Provide timely updates and support regarding the status of returns and refunds to keep customers informed and address concerns promptly.

Train Customer Service: Equip customer service representatives with the skills to manage return and refund inquiries efficiently and empathetically.

Monitor and Review: Regularly assess the return and refund process using customer feedback to identify and address any persistent issues.

2.2.3 Problem 03

2.2.3.1 Statement of the Problem

Chaldal.com faces significant vulnerabilities in digital security, exposing it to malware, viruses, and unauthorized access, which threatens customer data and operational integrity. There is an urgent need for robust security measures to protect against internal and external threats and to maintain user trust.

2.2.3.2 Summary of Findings

Chaldal.com faces significant security challenges, including malware infections, viruses, and unauthorized access. Issues with database operations also contribute to potential vulnerabilities. To address these, it is crucial to implement advanced security measures, enhance malware detection, and improve database protection.

2.2.3.3 Details of Findings

Chaldal.com faces several critical digital security threats that could compromise its e-commerce platform's integrity. Despite its robust design, the site is vulnerable to common security issues such as malware infections and unauthorized access by hackers. These threats pose a significant risk to sensitive customer data, potentially leading to data breaches and loss of trust. Additionally, the system encounters problems with database operations, including inaccurate read/write processes that can result in discrepancies in product and delivery information. These inconsistencies not only affect operational

efficiency but also heighten the risk of data manipulation and unauthorized access. Moreover, the platform's exposure to various operating systems used by customers increases the risk of malware introduction from diverse sources. This variability in user environments can lead to security vulnerabilities that are challenging to address with a one-size-fits-all approach. The current security measures are insufficient to counteract these complex threats, highlighting the urgent need for advanced protective strategies. Implementing sophisticated malware detection systems and enhancing overall security protocols is crucial to fortify the platform against these internal and external threats. Improving database security and ensuring comprehensive protection are essential steps in maintaining operational integrity and safeguarding customer data.

2.2.3.4 Recommendation

Enhance Network Security: Strengthen network defenses to prevent hacking attempts. Implement advanced firewalls, intrusion detection systems, and regular security audits to safeguard sensitive data.

Implement Robust Malware Detection: Deploy comprehensive malware detection systems compatible with various operating systems to identify and neutralize threats from multiple sources.

Improve Database Security: Address database read/write inconsistencies by ensuring accurate and secure database operations. Regularly update and maintain database security protocols to prevent data discrepancies and unauthorized access.

2.2.4 Problem 04

2.2.4.1 Statement of the Problem

E-commerce platforms often face challenges with dynamic pricing, where the same product may be priced differently across various sites due to frequent fluctuations. This variability can lead to confusion and distrust among customers, as well as operational difficulties for the business.

2.2.4.2 Summary of Findings

The implementation of a dynamic database is a practical solution to address fluctuating product prices. This system updates prices in real-time based on market conditions, ensuring accuracy and consistency. By adopting this approach, e-commerce platforms can improve customer trust, reduce pricing discrepancies, and stay competitive by reflecting current market trends.

2.2.4.3 Details of Findings

Supermarket item prices are frequently subject to change, complicating efforts to maintain consistent pricing and attract customers. This volatility can confuse shoppers and undermine brand loyalty, as consumers become increasingly price-sensitive. High operating costs further strain profitability, forcing retailers to reactively adjust prices to stay competitive. Additionally, fluctuating prices can create a perception of unpredictability, making it difficult for customers to establish trust in the pricing model of a supermarket.

Grocery delivery services aim to capture market share by offering appealing prices and added value. However, the plethora of alternatives makes customer retention challenging. Frequent price adjustments require constant system updates, leading to potential downtime that disrupts business operations and hinders customers from placing orders. Such interruptions can frustrate users and negatively impact overall sales and satisfaction. Moreover, inconsistent pricing can lead to customer frustration when they find discrepancies between advertised prices and final checkout totals.

To address these challenges, supermarkets need a robust pricing system that can manage dynamic pricing efficiently, minimizing operational disruptions. Implementing automated pricing algorithms can streamline adjustments based on market trends while maintaining clear communication with customers about price changes. This transparency helps mitigate confusion and enhances trust. By focusing on value and customer relationships, supermarkets can enhance loyalty and navigate the complexities of pricing fluctuations effectively. Furthermore, integrating customer feedback mechanisms can provide insights into pricing perceptions, allowing retailers to adjust strategies proactively. This comprehensive approach not only addresses the immediate challenges of dynamic pricing but

also fosters a stronger connection with the customer base, ultimately driving long-term success.

2.2.4.4 Recommendation

Implement Dynamic Pricing Database: Utilize a dynamic database to update prices in real-time based on market conditions, ensuring accurate and consistent pricing across the platform.

Enable Real-Time Pricing Updates: Develop features that allow real-time pricing updates without requiring the website or app to go offline, ensuring continuous operation and enhanced customer experience.

Offer Strategic Discounts: Introduce slight price reductions or promotional offers to attract customers and increase orders, thereby boosting website traffic and sales while maintaining competitiveness.

2.2.5 Problem 05

2.2.5.1 Statement of the Problem

Order delivery disruptions pose a significant challenge for e-commerce platforms, leading to customer dissatisfaction when faced with delays, incorrect deliveries, or fraudulent transactions. These issues can negatively impact the company's reputation and customer trust.

2.2.5.2 Summary of Findings

The absence of strict delivery deadlines on the website leads to problems such as expired products and damaged items during transport. Addressing these issues is essential to ensure customer satisfaction and maintain operational efficiency.

2.2.5.3 Details of Findings

The grocery delivery sector, which saw significant growth during the pandemic, continues to expand as consumer demand for convenience rises. E-commerce platforms like Chal-

dal.com are integral to this evolution, relying on effective online systems to manage the logistics of deliveries from warehouses or retail locations. These platforms not only facilitate customer orders but also streamline operations to meet the growing expectations of today's consumers.

A key factor in this complex landscape is the establishment of partnerships with local producers, which adds both opportunities and challenges. While collaborating with local suppliers can enhance product offerings and ensure freshness, it also necessitates efficient logistics to handle increased order volumes. As demand surges, the ability to manage these logistics effectively becomes critical. This includes coordinating delivery schedules, managing inventory levels, and ensuring that products are delivered promptly and in optimal condition.

Efficient logistics management is essential, particularly for covering large delivery areas while maintaining product quality. Proper staffing and vehicle management play a crucial role in safeguarding the integrity of products during transit. Grocery items, especially perishables, require careful handling and timely delivery to ensure they arrive in prime condition. The pressure to deliver fresh groceries quickly means that any inefficiencies in the system can lead to product spoilage and dissatisfied customers.

Last-mile delivery poses a significant challenge in this sector. Customers increasingly expect their orders to arrive within 30 to 40 minutes of placing them. This high expectation places immense pressure on delivery operations, and failing to meet these timelines can result in customer frustration, negative reviews, and ultimately, loss of business. Moreover, if delivery personnel are faced with an unreliable scheduling system, it can lead to confusion and miscommunication, further exacerbating delays and diminishing customer satisfaction. In conclusion, as the grocery delivery sector continues to evolve, companies like Chaldal.com must prioritize the enhancement of their logistics and operational frameworks.

2.2.5.4 Recommendation

Implement Distributed Server Systems: Adopt a distributed server system to manage high traffic and prevent performance issues or crashes, especially during peak times.

Optimize Last-Mile Delivery: Focus on improving last-mile delivery efficiency to en-

sure timely and accurate deliveries, meeting customer expectations.

Enhance Delivery Instructions: Provide clear instructions and establish robust processes for delivery personnel to prevent disruptions and ensure smooth operations.

Regularly Update Delivery Logistics: Continuously review and refine delivery logistics to address potential problems, maintain product quality, and enhance overall service reliability.

2.3 Conclusion

In conclusion, recognizing needs effectively establishes a solid foundation for subsequent system analysis stages, including planning, design, and implementation. This step is crucial for developing or improving systems that address identified challenges and meet stakeholder expectations. The analysis of Chaldal.com has revealed significant issues such as user experience problems, digital security threats, order delivery disruptions, dynamic pricing issues, and return and refund policy challenges. By addressing these concerns, the system analyst aims to provide targeted recommendations to enhance Chaldal's operations, improve user satisfaction, and strengthen its market position. This section has outlined the issues and proposed solutions to advance the system's development, aligning with both organizational goals and customer needs.

Chapter 3

Analysis

3.1 Introduction

Information gathering and detailed analysis play a crucial role in the system analysis process, providing a solid foundation for understanding the existing or proposed system and its requirements. This phase involves collecting, organising, and analysing relevant data and information to gain a comprehensive understanding of the system's context, objectives, constraints, and stakeholders' needs.

The primary objective of information gathering and detailed analysis is to identify the current problems or opportunities that necessitate the development or improvement of the system. It involves working closely with stakeholders, such as end-users, managers, subject matter experts, and IT personnel, to gather a wide range of perspectives and insights.

By conducting thorough information gathering and detailed analysis, system analysts can gain a deep understanding of the system's context and stakeholders' needs. This enables them to make informed decisions and propose appropriate solutions that align with the goals and requirements of the organisation or project.

3.2 Information Gathering

Gathering information is a crucial aspect of the feasibility analysis process. It involves both scientific and artistic elements. Scientifically, it requires structured methods and tools to ensure efficiency. Artistically, it demands mental agility to obtain optimal outcomes. No standard procedures are defined when it comes to gathering information. However, an important rule that must be followed is the following: information must be acquired accurately and methodically, under the right conditions and with minimum interruption to the individual from whom the information is sought. Some basic tools are:

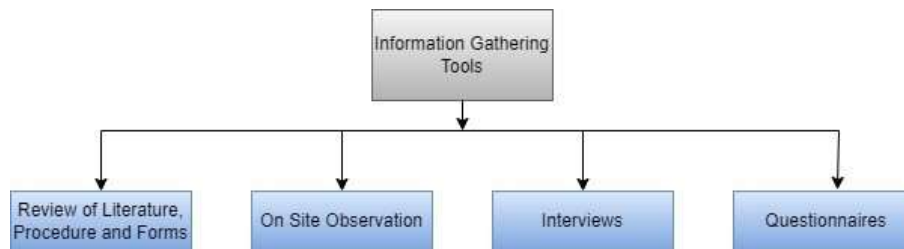


Figure 3.1: Information gathering tools

3.2.1 Review of Literature, Procedure and Forms

Review of Literature, Procedures, and Forms is a fundamental step in the information gathering process, as it provides a comprehensive understanding of existing knowledge, established practices, and standardized documents relevant to the system under analysis. This review helps system analysts build a strong foundation for further data collection and analysis.

Review of Literature

The review of literature involves examining industry standards, best practices, and research pertinent to the e-commerce and grocery delivery sectors. For Chaldal Ltd., this encompasses analyzing various sources to understand current trends and advancements:

- **Industry Standards:** Investigating industry standards for e-commerce and logistics helps in benchmarking Chaldal's practices against best practices in the field.

This includes understanding the latest developments in online grocery shopping, customer satisfaction metrics, and emerging technologies in supply chain management.

- **Best Practices:** Reviewing case studies and reports from successful e-commerce platforms provides insights into strategies that enhance operational efficiency and customer satisfaction. This includes exploring how other companies manage inventory, handle delivery logistics, and maintain high customer service standards.
- **Technology Advancements:** Analyzing recent research on technology advancements such as automated inventory systems, real-time tracking, and predictive analytics offers valuable information on tools and solutions that Chaldal could adopt to improve its operations.

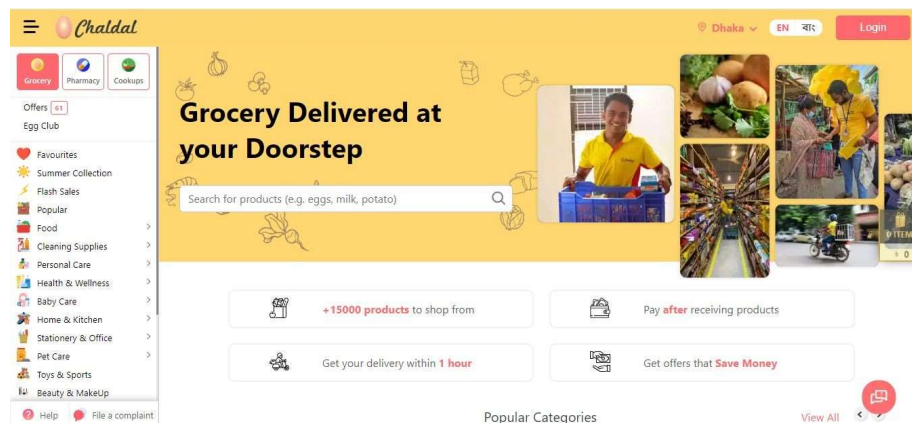


Figure 3.2: Homepage of Chaldal.com website

Review of Procedures

Reviewing internal procedures is essential to understanding how Chaldal's operations are managed and where improvements can be made:

- **Order Processing:** Evaluating the end-to-end process from when customers place orders on Chaldal's website or app to the final delivery. This includes analyzing the steps involved in order receipt, processing, and fulfillment to ensure efficiency and accuracy.

- **Inventory Management:** Examining how Chaldal manages its stock levels across multiple micro-warehouses. This includes procedures for restocking, handling inventory discrepancies, and ensuring stock availability to meet customer demands.
- **Delivery Operations:** Analyzing the logistics of product delivery from warehouses to customers. This includes assessing the coordination between warehouse staff and delivery personnel, and evaluating the effectiveness of delivery routes and schedules.
- **Customer Service:** Reviewing the procedures for addressing customer inquiries, complaints, and feedback. This ensures that customer issues are resolved efficiently and effectively, contributing to overall customer satisfaction.

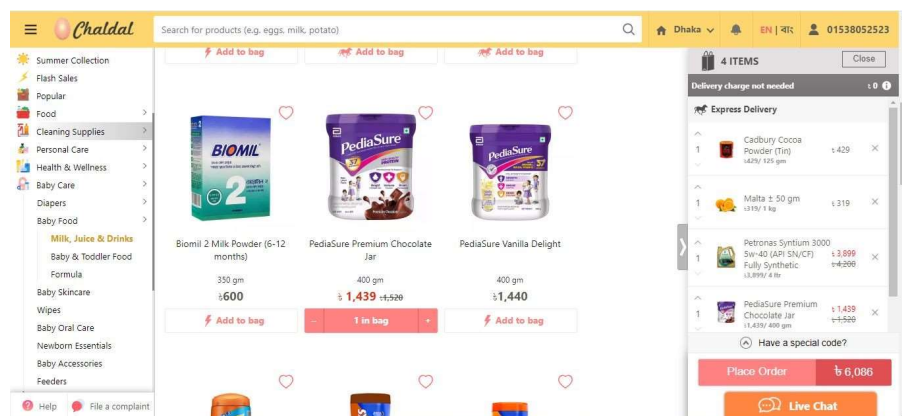


Figure 3.3: Place an Order process of Chaldal.com

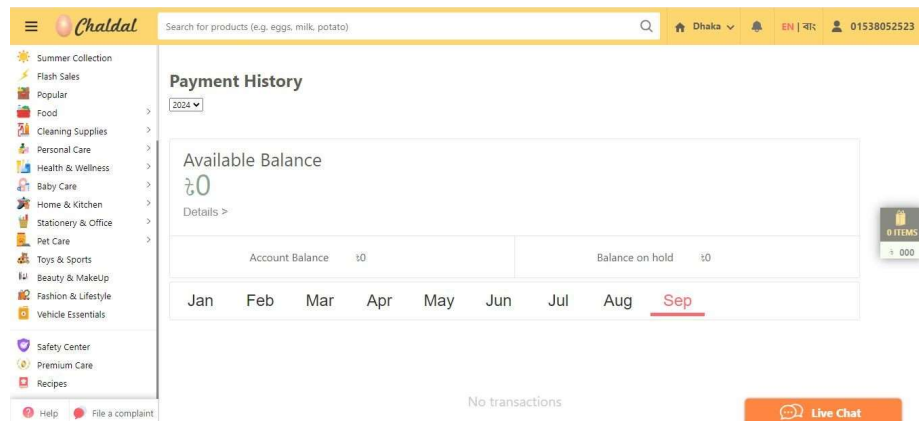


Figure 3.4: Payment History of Chaldal.com

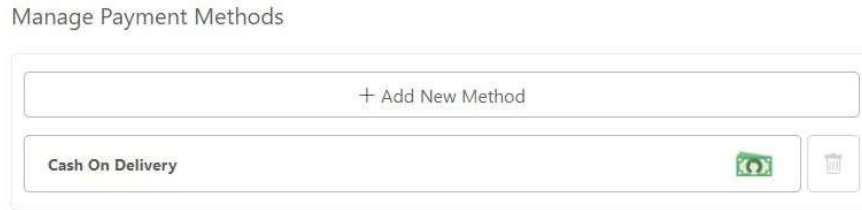


Figure 3.4: Payment Method of Chaldal.com

Review of Forms

Forms used by Chaldal Ltd., whether digital or paper-based, are integral to data collection and process management:

- **Order Forms:** Assessing the forms used by customers to place orders, focusing on their user-friendliness and accuracy in capturing order details. Ensuring that these forms facilitate a smooth ordering process is crucial for minimizing errors.
- **Inventory Forms:** Reviewing forms related to inventory tracking, including stock entry and exit logs. These forms are vital for maintaining accurate inventory records and managing stock levels effectively.
- **Delivery Forms:** Evaluating forms used by delivery personnel to confirm product receipt and delivery. These forms should be reviewed for completeness and effectiveness in tracking delivery status and managing delivery-related issues.
- **Customer Feedback Forms:** Analyzing forms used to gather customer feedback and complaints. This review helps in understanding customer concerns and identifying areas where service improvements are needed.

By thoroughly reviewing these forms, we ensure that Chaldal's data collection processes are robust and aligned with operational and customer service goals. Identifying gaps or issues in form design and usage helps in enhancing data accuracy and operational efficiency.

3.2.2 On Site Observation

On-site observation is a vital method for understanding the real-world operations of a system, providing insights that are often not captured through other data collection techniques. During my visit to Chaldal Ltd., I conducted thorough observations to gain a better understanding of their operational processes, workflow, and challenges. The primary objective of the on-site observation was to evaluate Chaldal's operational efficiency, particularly focusing on their delivery processes, warehouse management, and customer service interactions. The visit aimed to identify any inefficiencies or issues that may affect their performance and customer satisfaction.

Observations Made

1. **Warehouse Operations:** The warehouse is efficiently organized with a clear layout that supports effective picking and packing. Advanced inventory management systems are in use, facilitating stock tracking and demand prediction. However, occasional delays in item retrieval were noted, especially during peak order volumes.
2. **Delivery Process:** Delivery operations are well-coordinated, with dedicated teams adhering to strict schedules. Real-time tracking systems enhance transparency but occasionally face glitches. Delivery routes sometimes experience congestion, leading to minor delays.
3. **Customer Service:** The customer service team handles inquiries through both phone and digital channels. While responsive, there were some complaints about late or incorrect deliveries. Additional resources could help manage peak times more effectively.
4. **Technology and Systems:** Chaldal uses automated systems for inventory and order management, though some system lag was observed during high traffic periods.
5. **Staff Interaction:** Staff demonstrated professionalism and a strong understanding of their roles. Feedback suggests a need for additional training on new systems and technologies.

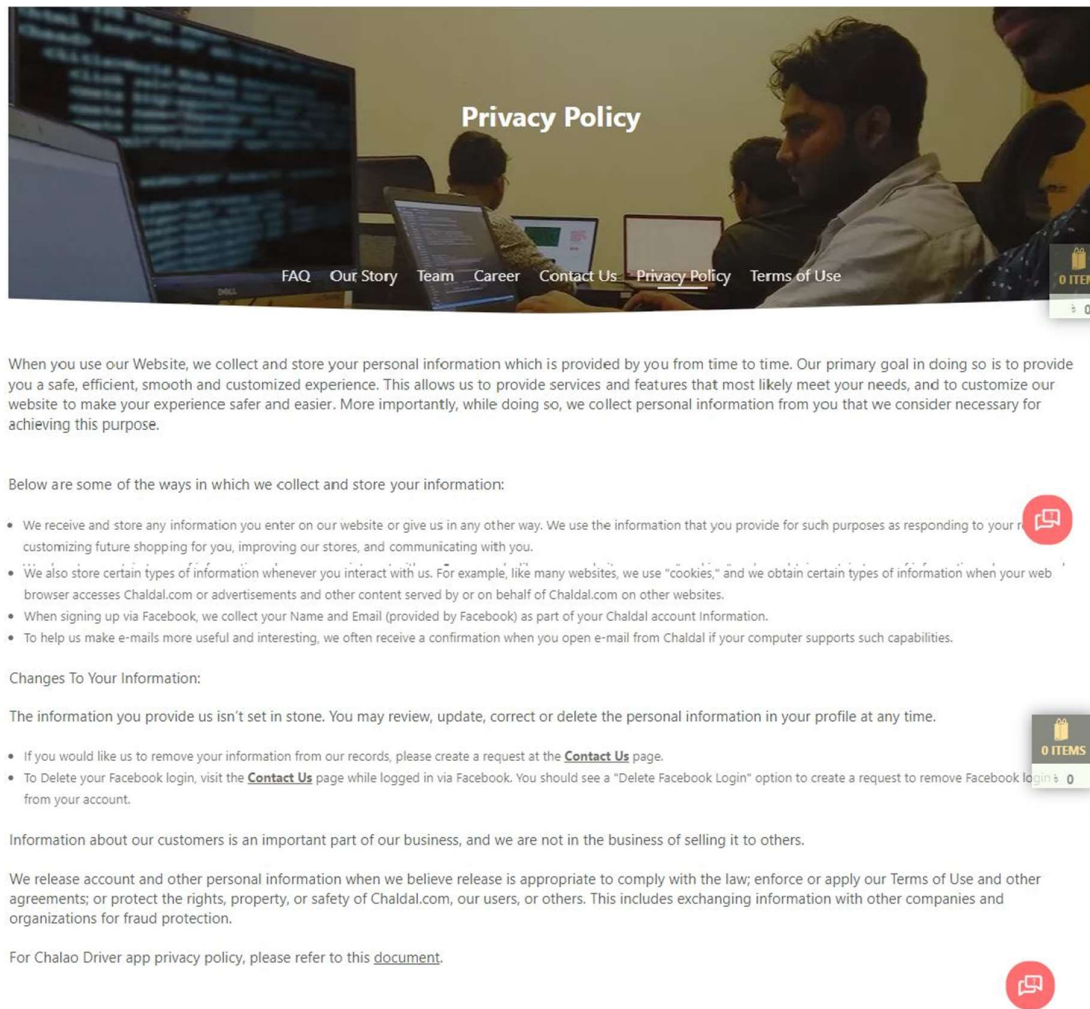


Figure 3.3: Privacy Policy of Chaldal company

Motto: Save money, Save time

Mission: Their mission is to have a resourceful stock of low priced quality product and deliver it within 1 hour by your doorstep.

Vision: The vision is to be the best online and offline grocery store and earn customer satisfaction.

3.2.3 Interviews

Interviews are an important method in system analysis for collecting information directly from stakeholders and experts. They allow for direct interaction, enabling specific questions to be asked and detailed insights to be gathered about the system and its requirements. Depending on the project's scope, interviews can be conducted individually

or in groups.

Interview with Waseem Alim, Co-founder and CEO, Chaldal

1. Question: What do you feel is the biggest strength of our company right now?

Answer: Our company's greatest strength lies in the alignment of our team members, resulting in fewer conflicts and more productive discussions. Everyone works together with strong discipline, allowing us to break down problems into manageable steps and tackle them efficiently and smoothly.

2. Question: What was the path you took to get to where you are today?

Answer: The journey wasn't straightforward; it had many ups and downs. Without the dedication and belief of the co-founders and team members, we wouldn't have achieved what we have today. Initially, I faced numerous challenges with different aspects of the software. It was difficult to streamline the process, as certain parts of the website, like the payment system and the cart option, were highly sensitive and required careful handling.

3. Question: What goal do you have for the company?

Answer: Our focus is on continuous innovation and upholding strong values, which has led to improved employee retention and positive GMV growth. I've set a goal to expand the company to 1,000 members within the next 2-3 years, while also increasing our contribution to the country's GDP. Additionally, we aim to enhance the website's efficiency to better serve all end users.

4. Question: Why did you start (or want to be CEO of) this company?

Answer: I began this journey because I disliked the traditional "Boss" mentality, where the boss controls and dominates others. A true leader, in my view, nurtures their team and allows individuals to grow in their own unique ways. I've always strived to break away from conventional norms and embody true leadership. My goal has always been to guide the team toward efficient solutions. Becoming a CEO has been my aspiration since my undergraduate days, and I've nurtured this dream ever since. Over time, I've honed various technical skills to prepare myself for this

role.

5. Question: What have been the biggest challenges you've had to overcome?

Answer: I've encountered numerous challenges with my websites, including delivering consistent service, handling payment issues from various gateways, and dealing with server crashes. These have been the main drawbacks so far, but I've worked hard to resolve them. My technical team has been a constant support, and through these challenges, I've learned a great deal.

6. Question: How many people are working at present in the organisation?

Answer: My company has over 2500+ employees all over the country and has 29 retail warehouses.

7. Question: Are you and your managers having 1 on 1s with your staff?

Answer: As the CEO, I believe strong relationships between managers and staff are crucial, and while 1-on-1s are important for coaching and problem-solving, our focus on fostering a fun and community-like workspace is equally vital. Even with increasing workloads, we prioritize creating an environment where learning, growth, and camaraderie thrive.

8. Question: What is the current statement on Chaldal's state of operation?

Answer: We are expanding to new cities like Narayanganj and Chittagong, focusing on B2B and partnerships such as with the WFP at the Rohingya Camp. Key priorities include improving our supply chain, ensuring 99.9% service quality, and sustainable growth that emphasizes quality over discounts.

9. Question: What is the say on sustainable growth as a priority?

Answer: Sustainable growth means each order must be financially viable without investor reliance, covering salaries and technology investments. In Narayanganj, we've surpassed expectations with over 300 daily orders and aim for 1,000 monthly, focusing on high-quality service.

10. Question: What is most important to your organisation—mission, vision or values?

Answer: Our singular vision is to build an infinite organization where collaboration

and innovation thrive. We view competitors as inspirations for improvement rather than rivals.

11. Question: What is your say on Chaldal Payment Systems and Chaldal's ambition with payments?

Answer: We aim to develop a payment system akin to PayPal, pending Bangladesh Bank approval. This initiative will require significant responsibility and infrastructure investment, taking time to mature.

12. Question: What is the best and the worst part of being a CEO?

Answer: The best part is witnessing the growth of a talented team; the worst is recognizing that it won't last forever.

13. Question: What are your plans on priorities for 2024?

Answer: We aim to invest our time and resources wisely, focusing on one or two key opportunities rather than spreading ourselves too thin.

14. Question: Definitely, it will help you increase your influence and to some extent give you some control?

Answer: We prioritize empowerment over control, envisioning a neutral platform that fosters collaboration and attracts talented individuals, recognizing the importance of making the right strategic decisions for long-term success.

15. Question: What is the one thing you would change about the company?

Answer: To build the belief among the members so that all can look beyond what they already have and what they are now. Because improvement is limitless.

Interview with HR and the Admin

1. Question: What do you want to say about your role in Chaldal.com? **Answer:**

We handle recruitment, employee relations, payroll, and coordinate administrative functions.

2. Question: Tell me something about your experience here?

Answer: I enjoy working with a cooperative team and appreciate the effective problem-solving environment.

3. **Question:** Can you say something about your company's services?

Answer: Chaldal offers a wide range of household and consumer products through its online platform.

4. **Question:** What do you like most and least about your present position?

Answer: I love working with tech enthusiasts and face new challenges, with no dislikes.

5. **Question:** How do you participate in planning and development?

Answer: Challenges include attracting qualified candidates and ensuring a fair recruitment process.

6. **Question:** What kind of difficulties do you face in finding and recruiting employees?

Answer: We face several challenges in recruitment, including attracting the right candidates due to limited options and unqualified applicants, making it difficult to find the best fit. Ensuring fairness is another issue, as unconscious biases can occur despite our efforts, though we are exploring 'blind hiring' to minimize this. Additionally, the recruitment process must be fast and efficient, but administrative tasks and strained relationships between recruiters and hiring managers can hinder progress and coordination.

7. **Question:** Have you any comments about the candidates you face when recruiting?

Answer: I have not commented about them. But I think one thing needs to be said. The overall skill sets of the maximum candidates are not satisfactory. Compared to the number of applications, there should be more qualified applicants. Also, skills don't match with the educational qualifications of the candidates often.

8. **Question:** How do you conduct disciplinary actions?

Answer: We have our own policies and code of conduct. We always monitor all employee's disciplinary acts. We have an anonymous complaint box. Anybody can complain or give their opinion in that box anonymously. Whenever there comes a complaint against someone we investigate this.

9. **Question:** What do you do to manage the working environment of the company?

Answer: We are always open to take suggestions. Periodically, we take employees'

feedback about the working environment of our company and any suggestions about making it better. We then conduct a meeting, discuss with our CEO and admins to implement the suggestions. If we get approval, we then try to implement that in our company. We also conduct company tours and bonuses for good works such that none feels bored about their work. Thus, it is ensured that our employees always have a good working environment culture.

10. **Question:** Can you express your overall opinion about this company?

Answer: Chaldal.com is a very renowned Ecommerce based platform in Bangladesh. The working environment here is very friendly and full of enthusiasm for employees. I like workplaces where there is an emphasis on both teamwork and autonomy. Collaborating with others, exchanging ideas, flexibility to work, etc. should be a must for any workplace I think.

Questions from the Customer Section

1. **Question:** The contact number?

Answer: You can always call 16710 or mail us at support@chaldal.com.

2. **Question:** What are your delivery hours?

Answer: We deliver from 7:30 am to 11 pm every day. You can choose from available slots to find something that is convenient for you.

3. **Question:** How do I know when my order is here?

Answer: We accept cash on delivery and we also have Online Credit Card and Online bKash service. Don't worry, our Chaldal representatives should always carry enough change.

4. **Question:** I can't find the product I am looking for. What should I do?

Answer: We are always open to new suggestions and will add an item to our inventory just for you. There is a "Product Request" feature that you can use to inform us of your requirement. You can also call 16710 or mail us at support@chaldal.com to add an item to our inventory.

5. **Question:** What is your policy on refunds?

Answer: We offer a refund or return policy of seven (7) days on most unopened or unspoilt packaged products.

- For perishable products such as milk, fruits, and fresh vegetables, we have a one (1) day return policy.
- Diaper items must be returned for refunds before 10% or 3 pieces (whichever comes first) have been used.
- Certain products; Face Mask, Disposable Vinyl Gloves, Alcohol Pads, and Covid Testing Kits are not acceptable for refund or return either opened or unopened.

6. **Question:** What is your discounting policy?

Answer: We try to provide the best deals in Dhaka and many of our products are already discounted below the maximum retail price (MRP). We also offer discount codes under special circumstances, which are applied on the MRP. On any given product, we can only apply one type of discount. We always consider the best discount available to the customer. For example: If the MRP of a product is Tk. 100 and our list price is Tk. 92 – the product is already sold at an 8% discount. This means that if the user applies a discount code for a 5% discount, we will still consider the best discount available to the user and sell the product at Tk. 92.

7. **Question:** How much do deliveries cost?

Answer: There is a BDT 19 delivery fee for every order.

8. **Question:** What about quality?

Answer: We try our best to source the best quality items for you but if you are dissatisfied, you can always send them back to the delivery person. If you forget to do that, you can call us within 3 days and we will replace the item for free.

9. **Question:** How do I recharge?

Answer: To recharge your Credit, please select “Shopping Credit” and then “Recharge”. We support debit cards, credit cards, bKash, and internet banking. You will be unable to place your order if you do not have enough credit.

10. **Question:** Can I have multiple addresses under one account?

Answer: No, you cannot. One e-mail address and one contact number at a time can only be associated with one Chaldal Daily account.

11. **Question:** How do I change my delivery address?

Answer: To change your address, please contact our customer service through live chat. We will only change your delivery address if you are able to verify a change of address.

12. **Question:** I have an issue with my order. What should I do?

Answer: Please contact us through live chat and let us know your order number and issue.

13. **Question:** What if the item is out of stock?

Answer: We try to only list items that we can ensure stock availability of. If an item is out of stock, we will restock it within 2-3 days.

3.2.4 Questionnaires

From CEO Yes or No (Dichotomous) Questions:

1. Do you believe Chaldal.com has the potential to expand internationally? (Yes/No)
2. Is customer satisfaction the top priority at Chaldal.com? (Yes/No)
3. Are you satisfied with the current growth rate of Chaldal.com? (Yes/No)
4. Does Chaldal.com invest enough in employee training and development? (Yes/No)
5. Is the current product delivery system at Chaldal.com fully optimized? (Yes/No)

From Employee of Chaldal Company: Ranking Scale Questions (1 = Lowest, 5 = Highest)

1. Rank your satisfaction with the employee benefits offered by Chaldal.com. (1 to 5)
2. How would you rank the work-life balance at Chaldal.com? (1 to 5)
3. Rank the clarity of communication between management and employees. (1 to 5)

4. How would you rate the effectiveness of the training provided at Chaldal.com? (1 to 5)
5. Rank your overall job satisfaction at Chaldal.com. (1 to 5)

From Employee Rating Scale Questions (1 = Strongly Disagree, 5 = Strongly Agree)

1. Chaldal.com offers opportunities for professional development. (1 to 5)
2. I feel motivated to perform my best at Chaldal.com. (1 to 5)
3. I have a good relationship with my team members. (1 to 5)
4. The feedback I receive from my manager helps me improve my performance. (1 to 5)
5. I am satisfied with the salary and compensation package at Chaldal.com. (1 to 5)

From User Fill-in-the-Blank Questions

1. I use Chaldal.com for ordering groceries _____ times per week.
2. The feature I use most often on the Chaldal app is _____.
3. My last order from Chaldal.com arrived within _____ hours.
4. The payment method I prefer to use on Chaldal.com is _____.
5. The reason I started using Chaldal.com was _____.

From User Multiple Choice Questions (MCQ)

1. How often do you place an order on Chaldal.com?
 - (a) Daily
 - (b) Weekly
 - (c) Monthly
 - (d) Occasionally

2. What is your primary reason for using Chaldal.com?
 - (a) Convenience
 - (b) Prices
 - (c) Product variety
 - (d) Promotions and discounts
3. Which device do you primarily use to place orders on Chaldal.com?
 - (a) Mobile App
 - (b) Website
 - (c) Phone Call
 - (d) In-store Kiosk
4. How long does it typically take for your order to be delivered by Chaldal.com?
 - (a) Less than 2 hours
 - (b) 2-4 hours
 - (c) 4-6 hours
 - (d) More than 6 hours
5. How do you find the pricing on Chaldal.com compared to traditional grocery stores?
 - (a) Much lower
 - (b) Slightly lower
 - (c) About the same
 - (d) Higher

3.3 Information Representation/Analysis

3.3.1 Plot various charts and graphs of the above mentioned

1. Are you satisfied with your work space's environment?

• 1 (Not Satisfied) • 2 • 3 • 4 • 5 (Very satisfied)

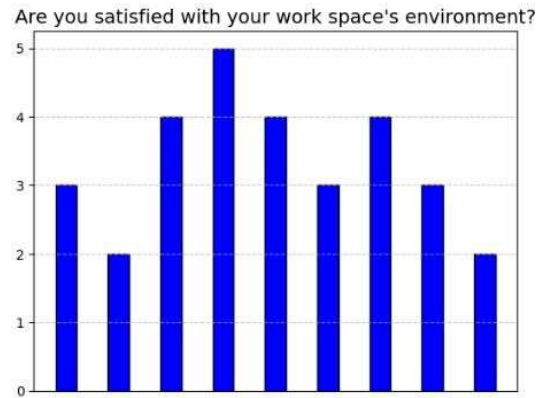


Figure 3.4: Satisfaction with work environment

2.How do you cope up with totally new technologies used in newly developed products and services?

- Helped by advisor
- By self-learning
- Helped by team-mates and senior employees

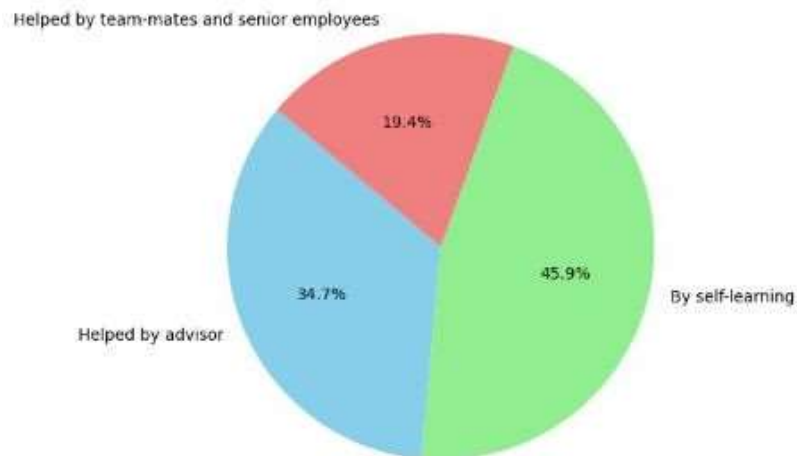


Figure 3.5: Adapting Strategies for Embracing Technological Changes

3.Does this job help you to achieve your goal?

- Yes
- No
- Maybe

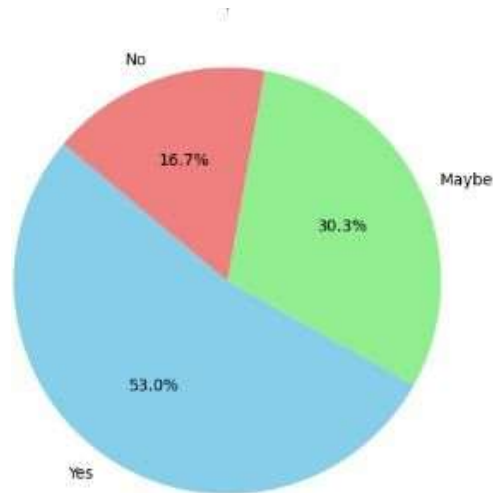


Figure 3.6: Distribution of Key Job Responsibilities Contributing to Career Goals

4. How do you feel transparent about the management?

- Strongly
- Don't Know
- Not at all

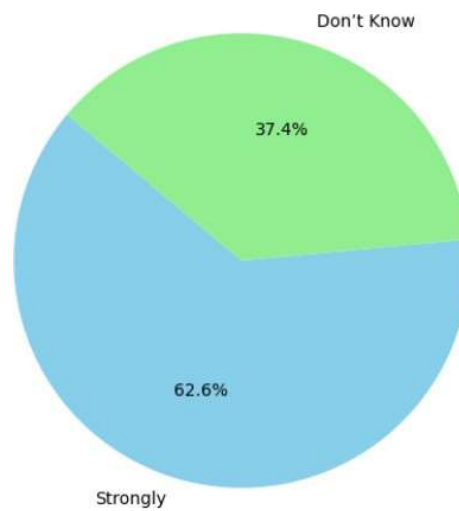
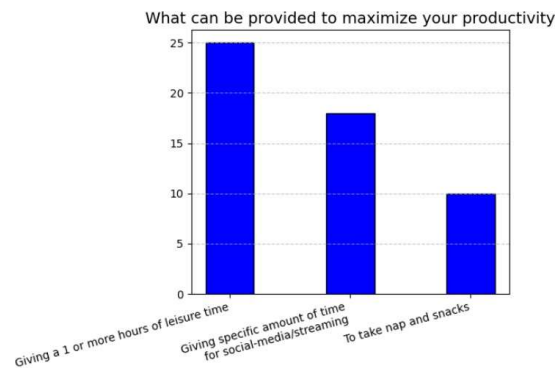


Figure 3.7: Perceptions of Transparency in Management Practices

5. What should a company provide you to boost your capability in office?

- Giving a 1 or more hours of leisure time
- Giving certain amount of time for social-media streaming
- To take a nap and snacks



Questionnaires for App Users

1. How many times have you brought any product from Chaldal?

- Never
- Once
- More than once

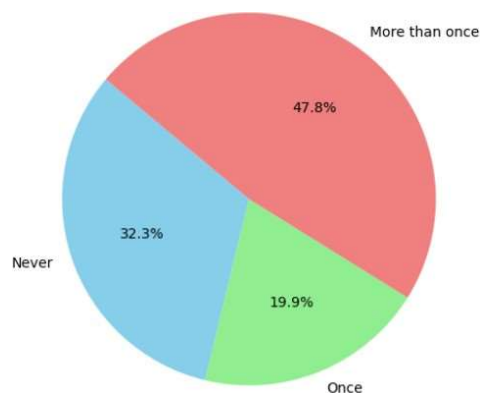


Figure 3.9: Frequency of Purchases from Chaldal

2. Do you like the product collection on the website of Chaldal?

- Yes
- No

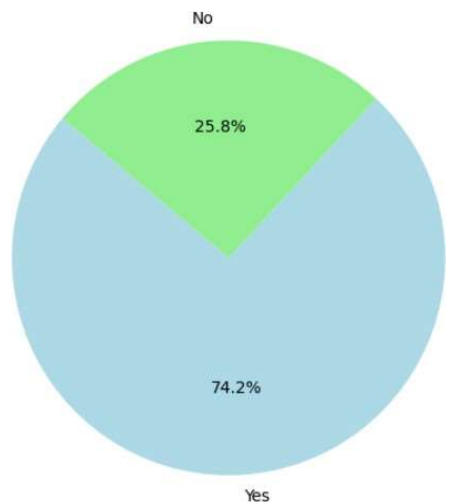


Figure 3.10: DSatisfaction with Chaldal’s Product Collection

4. Have you found your desired product from Chaldal?

- Yes
- No

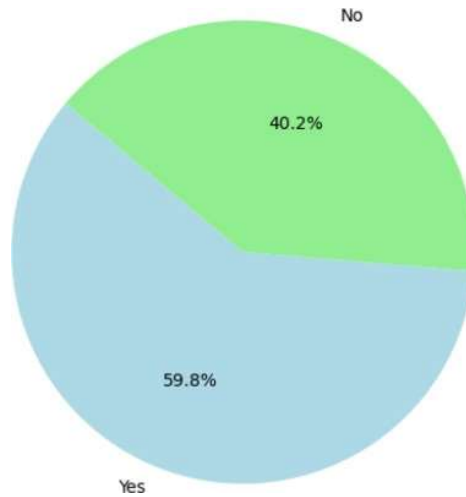


Figure 3.12: Success in Finding Desired Products on Chaldal

5. How is the product quality?

- Good
- Excellent
- Not so good

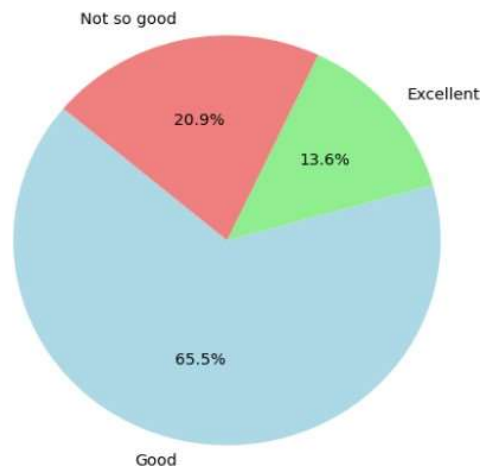


Figure 3.13: Assessment of Product Quality from Chaldal

6. What is the price range on the website?

- High
- Normal
- Low

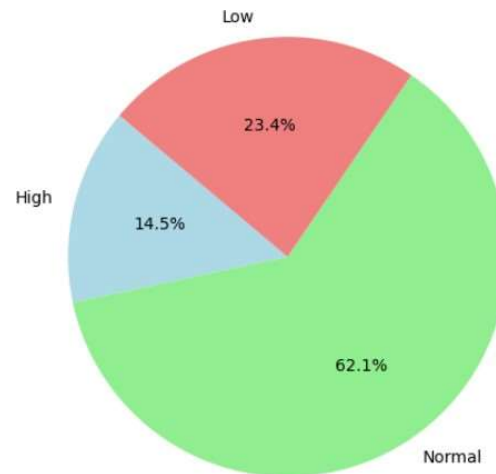


Figure 3.14: Perceptions of Chaldal's Price Range

3.3.2 DFD of Existing System

DFD is also known as the “Bubble Chart.” Its two main objectives are to specify system requirements and to identify significant modifications that will become programs in system design. At the start of the design phase, the needs and specifications are functionally broken down into their component elements. A DFD shows the system's data flows and the corresponding data transformations as lines and bubbles.

The provided Data Flow Diagram (DFD) illustrates (Fig:3.15) the workflow of Chaldal.com's order management system, detailing the steps involved in processing customer orders and managing inventory.

The DFD begins with the “Customer” process, where customers place their orders by providing necessary details such as product selection and payment method. This information flows to the “Edit Order” process, where order details are verified and edited as needed. If the product is available, the system proceeds to the “Payment” process.

In the “Payment” process, the customer pays for the order, and the system sends an invoice to the customer while verifying the payment. Once payment is confirmed, the flow transitions to the “Warehouse” process, which checks the availability of stock and retrieves product information from the inventory.

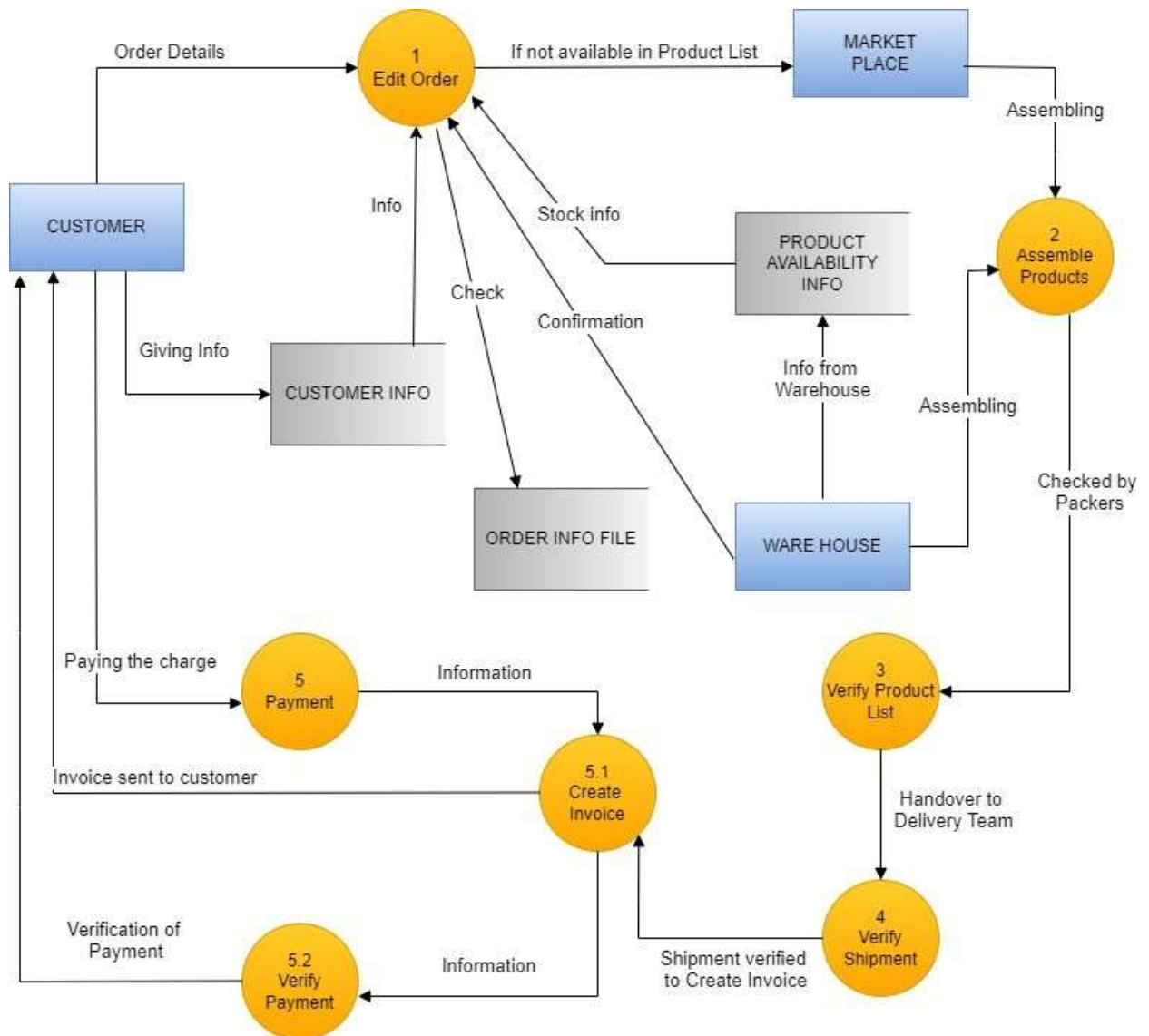


Figure 3.15: DFD for existing system of Chaldal.com

Next, the “Assemble Products” process takes place, where warehouse staff prepare the products for delivery. This is followed by the “Verify Product List” process, where the assembled products are checked for accuracy before shipment. The final step involves the “Verify Shipment” process, ensuring that the delivery team has the correct items for distribution.

After successful verification, the products are handed over to the delivery team for shipment to the customer. Throughout this process, the DFD highlights key interactions and flows of information, ensuring a streamlined experience for customers and efficient management of the supply chain.

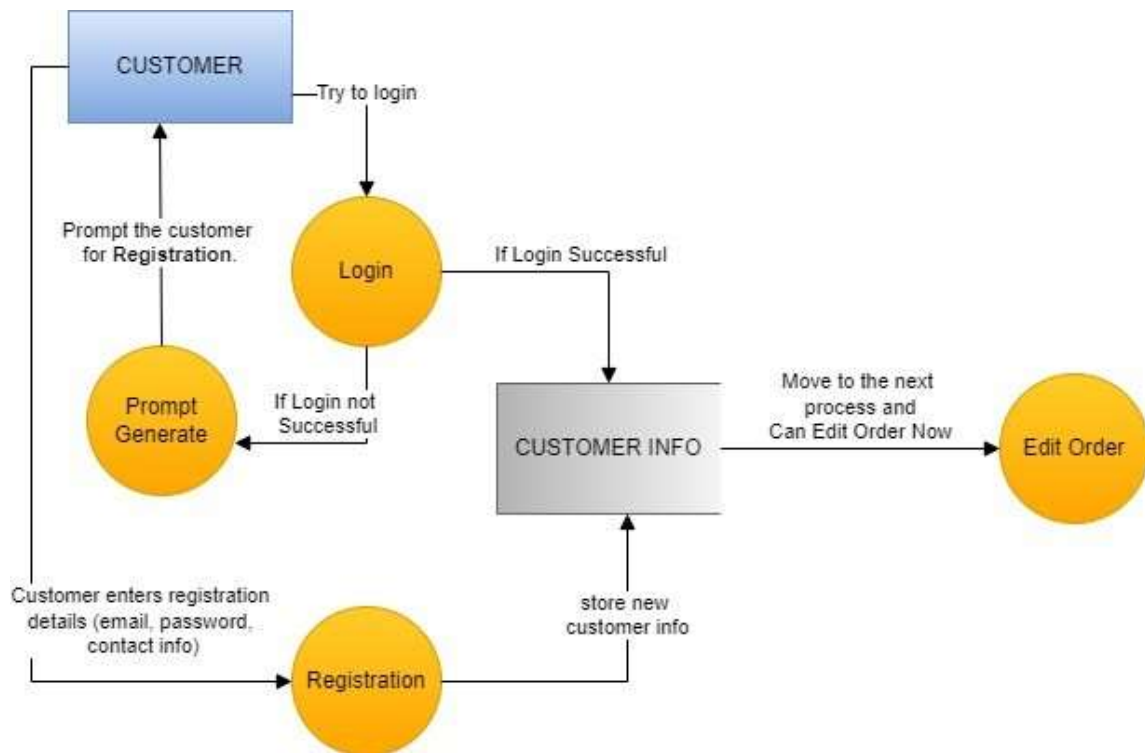


Figure 3.15.1: Login and registration DFD for existing system of Chaldal.com

Finally, the system concludes with the “Delivery Confirmation”, where customers receive their orders, and the overall process is evaluated for any necessary improvements, including potential adjustments in stock predictions and fraud detection measures to enhance service quality and reliability.

Additionally, Fig-3.15.1 presents the DFD for Customer Login and Registration, a pre-order step ensuring secure access. Customers either log in or register if new. Upon successful login or registration, customer data is retrieved to proceed with order edits.

Fig-3.15.2 outlines the DFD for Customer Support and Reports & Analytics. Customer Support resolves issues by accessing order and stock data, ensuring timely assistance. Reports & Analytics pulls data from key system files, generating insights to enhance operational efficiency and support strategic decisions.

This comprehensive DFD captures the essence of Chaldal.com’s operations, reflecting the interconnected processes that facilitate efficient order management and customer satisfaction.

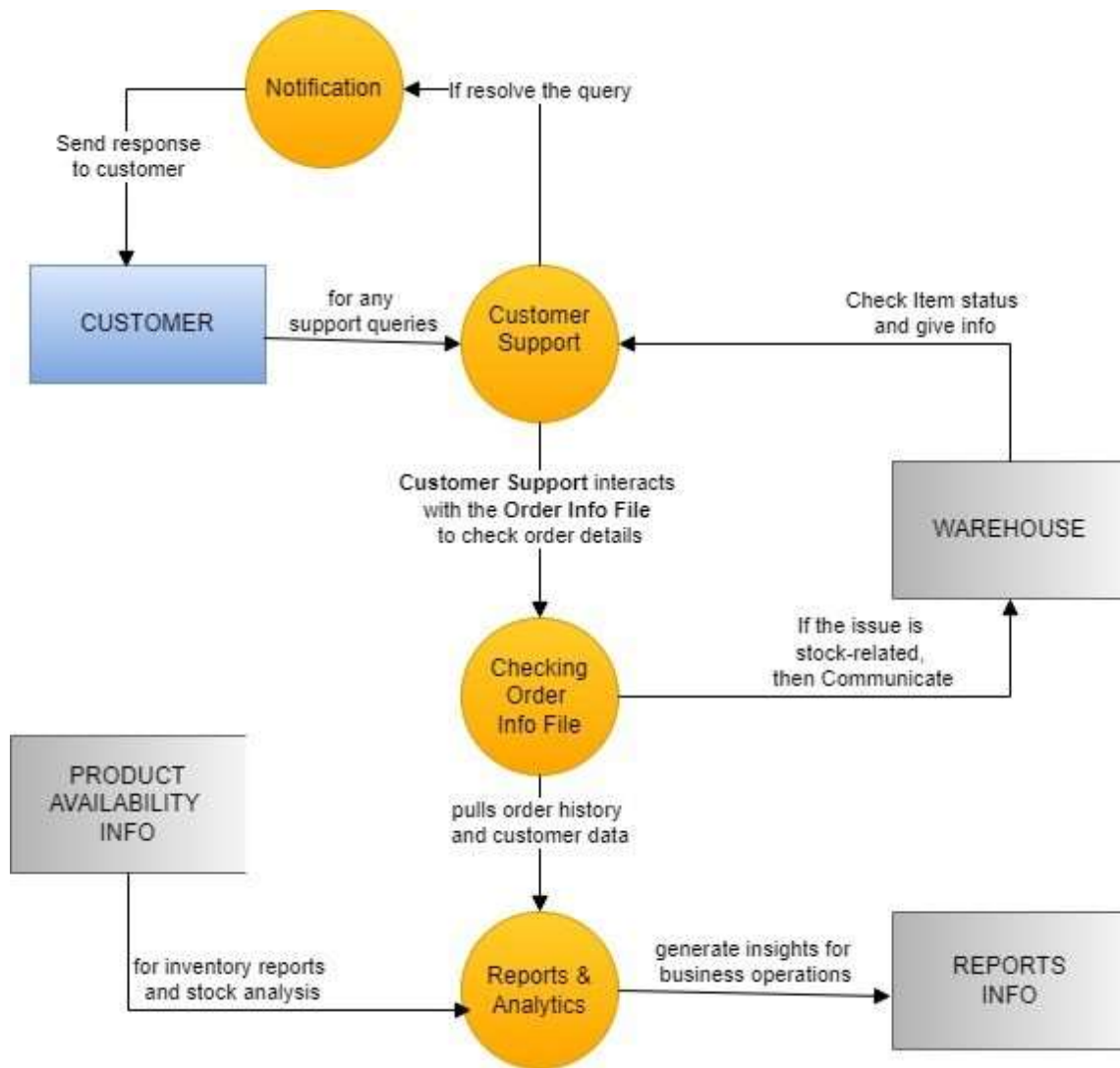


Figure 3.15.2: Customer support and Report analytic DFD for existing system

3.4 Cost-Benefit Analysis

We will now conduct a comprehensive cost-benefit analysis to evaluate the financial feasibility and overall value of the proposed system. This analysis aims to compare the total costs associated with implementing the system against the anticipated benefits and savings. By systematically assessing various expense categories, including hardware, personnel, facility, operating, and supply costs, we can determine the economic viability of the project. This approach will help us identify potential cost savings, efficiencies, and return on investment, providing a clear understanding of whether the benefits justify the expenditures. The results of this analysis will guide decision-making and ensure that

resources are allocated effectively to achieve the desired outcomes.

3.4.1 Procedure for Cost-Benefit Analysis

Before starting a new project or building a plant, managers conduct a cost-benefit analysis to evaluate potential expenses and revenues. System development is an investment with costs throughout its lifecycle and benefits like reduced operating costs, improved corporate image, increased efficiency, or added revenue. Cost-benefit analysis helps determine if benefits outweigh costs, guiding decisions on project viability. The evaluation process includes specific steps outlined in Figure 3.4.1.

Types of Costs and Benefits

Understanding costs and benefits is crucial for Chaldal.com's system enhancements.

Tangible vs. Intangible Costs

Tangible costs are measurable, such as purchasing delivery vehicles or software. Intangible costs, like customer dissatisfaction due to delays, are harder to quantify but impactful.

Direct vs. Indirect Costs

Direct costs include specific expenses like new servers, while indirect costs, such as IT support, affect overall operations. Reducing indirect costs can yield savings.

Fixed vs. Variable Costs

Fixed costs, like vehicle purchases, remain constant, while variable costs, such as fuel and order processing, fluctuate with system usage.

Savings vs. Cost Advantages

Savings result from reduced operational costs, such as optimizing delivery routes. A cost advantage directly cuts expenses, like reducing fraud through online payments.

Annual Investment

Chaldal.com's annual investment includes both fixed and variable costs to ensure the system remains efficient and scalable over time. Fixed costs are one-time expenses, such as system upgrades, infrastructure enhancements, and purchasing necessary hardware. Recurring variable costs, on the other hand, depend on system usage, such as SMS verification services, server maintenance, and customer support resources.

The fixed costs ensure the system's foundational stability, while the variable costs scale with the company's growing operations. Regular assessment of these costs helps maintain financial feasibility and long-term sustainability, ensuring the system continues to deliver optimal performance without exceeding budgetary limits.

Cost Type	Item	Cost
Hardware Cost	Desktop Systems	\$3600
	Printers & Scanners	\$270
	Network Setup	\$900
	Data Servers	\$3150
	Backup Devices	\$270
	Total Expense	\$8190
Personnel Cost	Employee Salaries	\$20,000
	Performance Bonuses	\$7000
	Professional Training	\$3900
	Insurance Benefits	\$5670
	Shift Allowances	\$1500
	Total Expense	\$38,070
Facility Cost	Office Lease	\$21,600
	Utility Bills	\$5080
	Security Services	\$3240
	Office Renovations	\$18000
	Equipment Upkeep	\$360
	Total Expense	\$48,280
Operating Cost	Refreshments	\$2430
	Conference Expenses	\$2700
	Office Maintenance	\$2160
	Stationery	\$430
	Sanitation Services	\$2200
	Total Expense	\$9920
Supply Cost	Raw Materials	\$81,000
	Inventory Management	\$8500
	Packaging Supplies	\$21,600
	Shipping Costs	\$50,000
	Storage Fees	\$13,500
	Total Expense	\$174,600

Table 3.4.1: Several cost of the existing system

For this system: Fixed Costs could include infrastructure upgrades, software development, and integration of new technologies, such as the SMS verification system, estimated at

around \$1,000,000.

Variable Costs might involve ongoing expenses like SMS charges for notifications, server maintenance, or scaling operations, calculated as 5% of total revenue or transaction volume (e.g., \$2,000,000), resulting in \$100,000.

Annual Costs would include both fixed and variable expenses, ensuring the system's smooth operation and scalability, estimated at \$1,00,000 annually.

For this system:

$$\text{Fixed Cost} = \$1,000,000$$

$$\text{Variable Cost} = 0.05 \times 2,000,000$$

$$\text{Annual Cost} = \$100,000$$

Total Annual Investment is calculated as:

$$\text{Total Annual Investment} = \text{Fixed Cost} + (\text{Variable Cost} \times \text{Variable Product Amount}) + \text{Annual Cost}$$

$$= \$1,000,000 + (0.05 \times 2,000,000) + \$100,000$$

$$= \$1,000,000 + \$100,000 + \$100,000$$

$$= \$1,200,000$$

3.4.2 Break-Even Point Analysis

The break-even point is a critical metric where a business neither earns profit nor incurs a loss. It occurs when the total revenue generated matches the total costs, allowing the company to cover its fixed expenses. This analysis helps determine the sales volume or revenue required to offset all costs, making it an invaluable quantitative tool for managers. By understanding the break-even point, managers can gain insights into whether the income from a product or service is sufficient to cover its production costs, aiding in strategic decision-making such as pricing, competitive bidding, and financial planning.

Steps in Break-Even Analysis:

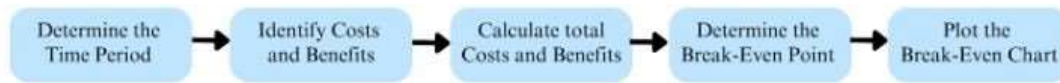


Figure 3.4.2: Steps in Break-Even Analysis

Break-Even Point Calculation

Given the following data:

Fixed Costs (F) = 30,000,

Variable Costs per Unit (V) = 11.75,

Selling Price per Unit (P) = 15.

The break-even point in units (BEP_u) is calculated using the formula:

$$\text{BEP}_u = \frac{F}{P - V}$$

Substituting the given values results in the following expression:

$$\begin{aligned}
 \text{BEP}_u &= \frac{30,000}{15 - 11.75} \\
 &= \frac{30,000}{3.25} \\
 &= 9,230.77 \text{ units.}
 \end{aligned}$$

The calculated break-even point of 9,230 units indicates that Chaldal Limited needs to sell this number of units to fully cover its fixed costs of \$30,000, with each unit contributing to covering the variable costs. At this volume, the company's total revenue will precisely match its total costs, resulting in neither profit nor loss. This information is crucial for financial planning and strategy, as it helps the company understand the minimum sales volume required to avoid financial losses and assists in setting realistic sales targets and pricing strategies to achieve profitability.

Graphic Result

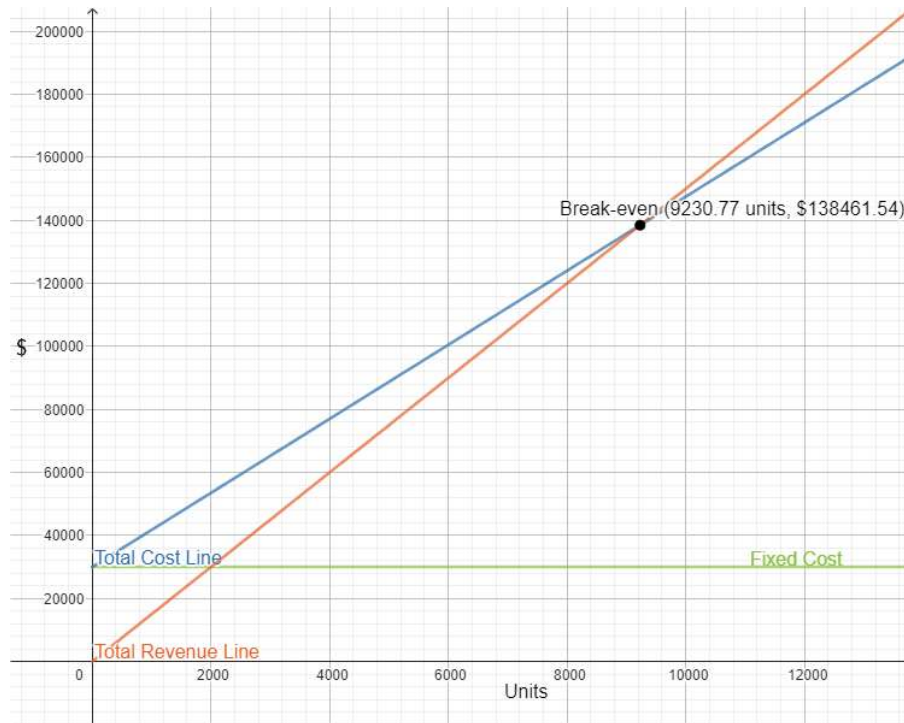


Figure 3.4.3: Break-Even Analysis

3.5 Detailed Feasibility Analysis

3.5.1 Root Cause Analysis of Identified Challenges

Chaldal.com faces several significant challenges that can be categorized into five primary areas: ineffective user interface and navigation, unclear return and refund policies, security vulnerabilities, issues with dynamic pricing, and disruptions in order delivery. Each of these problems significantly affects the platform's operational efficiency, customer satisfaction, and overall reputation. Addressing these challenges is crucial for the sustained growth and competitiveness of Chaldal.com in the e-commerce sector.

1. Ineffective User Interface and Navigation

Root Cause: The user interface and navigation system on Chaldal.com are currently not optimized, which complicates the process of locating products and completing pur-

chases. The website's complex layout and slow loading times hinder a seamless shopping experience.

Performance Impact: This inefficiency can lead to increased bounce rates as frustrated users abandon their shopping carts, resulting in lost sales opportunities and diminished customer engagement, ultimately impacting the platform's revenue.

Behavioral Impact: Customers who are used to more intuitive and responsive platforms may feel exasperated by the cumbersome navigation, which can lead to negative word-of-mouth and decreased loyalty to the Chaldal brand.

2. Unclear Return and Refund Policies

Root Cause: The return and refund policies at Chaldal.com lack consistency and transparency, leading to confusion among customers. The absence of clear procedures and slow refund processing times erodes trust in the platform.

Performance Impact: Unclear policies can increase customer complaints and dissatisfaction, harming the brand's reputation. This situation may also overload customer service teams, detracting from their ability to assist other customers efficiently.

Behavioral Impact: Potential buyers may hesitate to complete purchases if they are unsure about the return process, which can lead to abandoned shopping carts and lower conversion rates.

3. Security Vulnerabilities

Root Cause: Chaldal.com is exposed to significant security risks, including malware and unauthorized access due to insufficient digital security measures. These vulnerabilities threaten customer data integrity and the overall reliability of the platform.

Performance Impact: Security breaches can result in data loss and financial fraud, exposing Chaldal.com to legal liabilities and damaging customer trust. Such incidents can significantly tarnish the platform's reputation and financial health.

Operational Impact: The need to frequently address security issues can divert resources away from other crucial operational areas, hindering the platform's overall effectiveness and growth.

4. Dynamic Pricing Issues

Root Cause: Chaldal.com struggles with challenges related to dynamic pricing, where frequent price fluctuations create confusion among customers. Inconsistencies in pricing across various platforms and unclear communication about these changes exacerbate the problem.

Performance Impact: Unstable pricing can lead to a lack of trust from customers, adversely affecting their retention and loyalty. Customers may opt for competing platforms that offer more stable pricing strategies.

Behavioral Impact: Customers who encounter unpredictable pricing may feel frustrated and undervalued, leading to decreased satisfaction with the services provided by Chaldal.com.

5. Disruptions in Order Delivery

Root Cause: Order delivery disruptions represent a significant challenge for Chaldal.com, characterized by frequent delays, incorrect deliveries, and inadequate communication. These issues contribute to rising operational costs and inefficiencies in logistics.

Performance Impact: Inconsistent delivery performance can lead to missed sales opportunities and an increase in customer complaints, ultimately harming the platform's reputation.

Behavioral Impact: Customers who experience delivery issues are likely to lose confidence in Chaldal.com, resulting in lower repeat purchase rates and increased customer churn.

3.5.2 Candidate Systems

The current candidate system at Chaldal has been researched and analyzed. In Chapter Two, we discussed several issues with the existing system. After a comprehensive review, we have identified two potential replacement systems that could offer greater efficiency than the current one.

Candidate System-I

Chaldal.com faces significant delivery delays. One solution can be paying delivery drivers on a daily basis. This means that drivers who consistently deliver on time would receive a small bonus added to their monthly salary.

Fake orders or fraudulent customers are another major concern. Switching the payment method from cash on delivery to advance payment can help to avoid this issue. Also, implementing SMS verification for customer IDs, similar to online money transfer services, could further reduce fraud.

Both the Chaldal.com website and mobile app would benefit from the addition of a comparison tool, such as sorting options (e.g., price from low to high), to improve user experience. Also, it can offer discounts and flash sales.

Characteristics

- **Focus:** Strengthening order security, improving customer communication, and enhancing inventory management at Chaldal.com.
- **Technology Requirements:**
 - Development of a sophisticated fraud detection algorithm.
 - Integration of an SMS notification system.
 - Upgrades to inventory management software.

Candidate System-II

The proposed system could involve the programming team developing a solution to detect fraudulent orders, block them from future process, and prevent the creation of new accounts by those users.

An SMS notification system should be implemented to confirm orders. For example, after an order is placed, the customer could receive an SMS listing the items they ordered.

The stock prediction algorithm needs to be refined to accurately predict demand at the regional level.

Characteristics

- **Focus:** Strengthening order security, improving customer communication, and enhancing inventory management at Chaldal.com.
- **Technology Requirements:**
 - Development of a sophisticated fraud detection algorithm.
 - Integration of an SMS notification system.
 - Upgrades to inventory management software.

Identifying the Characteristics of Candidate Systems

We need to select the most suitable system from the candidate options we have evaluated. To do this, we used several criteria to assess which design was superior. We will first discuss the features of the two potential systems and then examine the qualitative evaluation matrix, which is presented in tables.

Features	Candidate System I	Candidate System II
Platform	Mobile app	App and website
SMS authentication	Yes	Yes
SMS notification	Yes	Yes
Purchase history track	No	Yes
Advance payment method	Yes	No
Delivery man's reward	Daily	Not available

Table 3.1: Characteristics for the proposed candidate systems

Performance and Cost-Effectiveness

The data does not clearly indicate the effectiveness of the potential systems based on their attributes. To better evaluate them, we will set specific criteria and present them in a qualitative evaluation matrix. These criteria will include factors such as user-friendliness, security, measures to prevent fraudulent orders, identity authentication, payment methods, SMS notifications, and development costs. We will assess both implementation and

design costs, as well as user education and training, system development, and operational costs. This evaluation aims to identify the most cost-effective solution.

Evaluation Criteria	Candidate System I	Candidate System II
Easy-to-use	Very good	Satisfactory
Data safety	Very good	Good
Fraudulent order	Very few	Few
Cost-effective	Good	Good
Authentication	Good	Satisfactory
Risk factors	Good	Satisfactory

Table 3.2: Qualitative evaluation Matrix

Evaluation Criteria	Candidate System I	Candidate System II
Easy-to-use	90-95%	70-80%
Data safety	80-85%	75-80%
Fraudulent order	30-40%	60-65%
Cost-effective	90-95%	70-75%
Authentication	80-90%	70-80%
Risk factors	85-95%	75-85%

Table 3.3: Cost evaluation matrix

Weighting System Performance and Cost Data

The above two matrices highlight the pros and cons of the proposed solutions. Based on the discussion, we can assess their efficiency by focusing on key criteria like user-friendliness, payment method, feedback system, and identity authentication. The first system is more user-friendly, as it works on both mobile apps and websites, while PCs only support the website. However, its payment method requires advance payment to prevent fake orders, aligning well with digital banking.

Evaluation Criteria	Weighting factor	Candidate System I		Candidate System II	
		Rating	Score	Rating	Score
Easy-to-use	5	5	25	4	20
Data safety	5	4	20	3	15
Fraudulent order	3	4	12	4	12
Cost effective	2	4	8	4	8
Authentication	3	3	9	3	9
Risk factors	4	4	16	3	12
			90		76

Table 3.4: Weighted evaluation matrix

Selecting the best candidate system

After conducting a thorough evaluation of the candidate systems, it becomes clear that the first proposed system exhibits a higher level of efficiency, particularly after addressing the identified challenges related to Chaldal.com. The comparative analysis presented in the tables reveals that the first system demonstrates significant advancements in several critical areas, including security protocols, payment processing mechanisms, and identity verification methods, all of which are superior to those found in the second system.

It is essential to recognize that no system is devoid of imperfections. Thus, the implementation of ongoing monitoring, comprehensive data collection, and meticulous analysis is paramount. These practices not only assist in identifying potential shortcomings but also play a vital role in achieving the overarching objectives of the organization. The process of system analysis functions as a strategic framework that involves a detailed examination of the system and its constituent components. This examination aids in clarifying goals, pinpointing inefficiencies, and outlining actionable improvements.

By employing a systematic problem-solving approach, we enhance the operational efficacy of the system, ensuring that each component functions cohesively towards fulfilling its designated purpose. Through rigorous data analysis, we can effectively confront emerging challenges and intricately understand the system's objectives. This deeper understanding

enables us to not only address immediate issues but also align the system's capabilities with the long-term strategic goals of the organization.

Moreover, fostering a culture of continuous improvement through iterative feedback loops can lead to sustained advancements in system performance. By regularly evaluating system performance metrics and user feedback, we can adapt and refine the system to meet evolving demands and technological advancements. This proactive approach not only mitigates the risk of obsolescence but also enhances user satisfaction and trust in the system.

In conclusion, the selection of the optimal candidate system is not merely a one-time decision but a dynamic process that requires ongoing attention and refinement. By prioritizing continuous monitoring and analysis, we can ensure that the chosen system not only meets current requirements but is also adaptable to future challenges, thereby maximizing its efficacy and longevity within the organizational framework.

3.6 Conclusion

The current system faced several inefficiencies that needed addressing. To enhance its performance, we proposed a new candidate system. A feasibility study was conducted to identify and select the most appropriate system. This study involved describing and evaluating potential systems, and choosing the best option based on various criteria. Key factors such as economic, technical, and behavioral considerations were integral to this analysis. We developed and assessed two candidate systems through multiple evaluation methods, ultimately determining which system would best meet our needs. A Data Flow Diagram was subsequently created to clearly illustrate the chosen system. Lastly, a comprehensive cost-benefit analysis was carried out to evaluate the advantages and costs associated with the proposed system.

Chapter 4

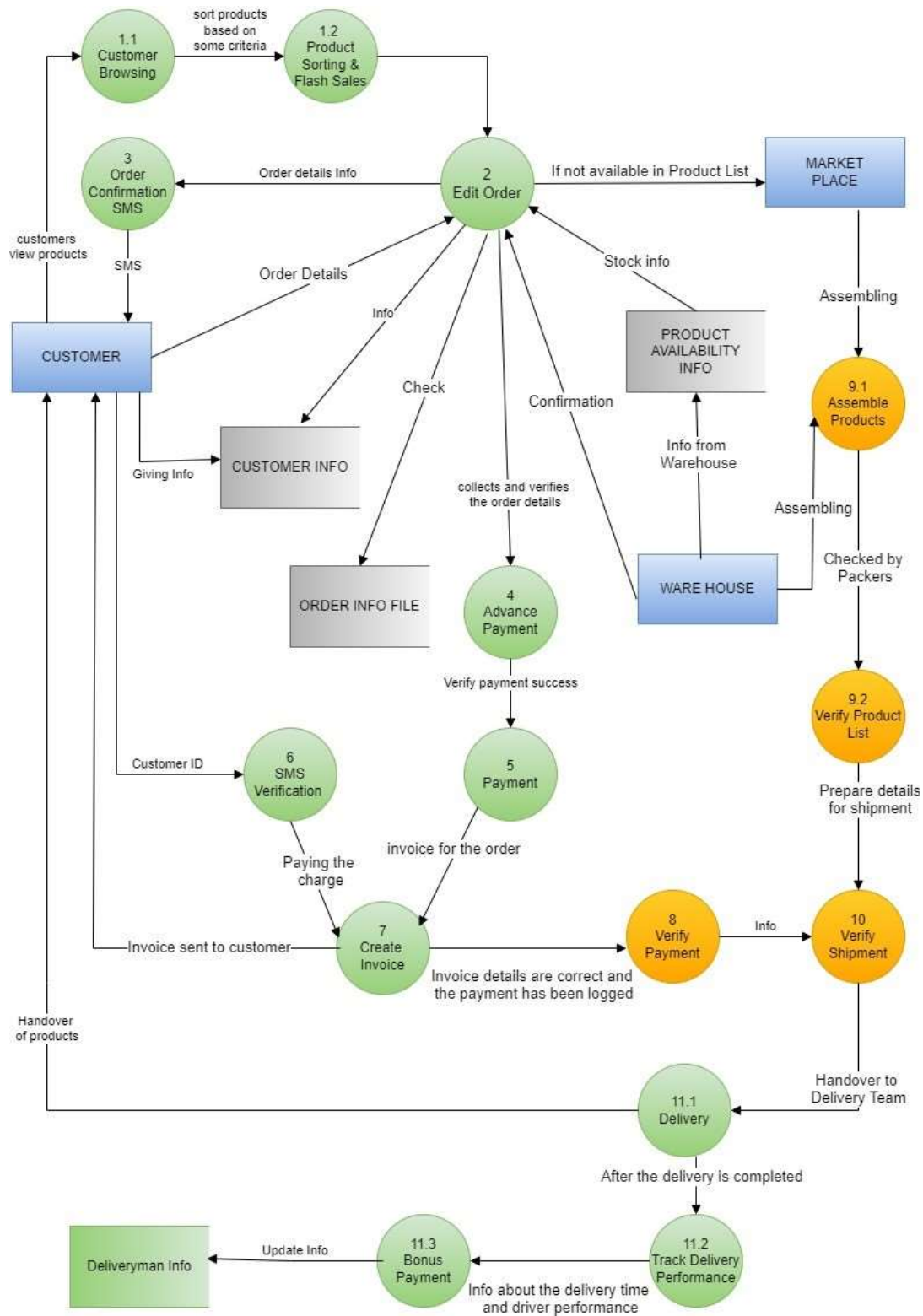
Design

4.1 Introduction

Systems design is the procedure of defining functions of a system like modules, architecture, diagram and their interfaces and data for a system based on the specialised requirements. It is the process of defining, developing and designing systems which ensures the specific needs and requirements of a business and organisation. Vital activities in system design and development include developing. System-level technical requirements and top-level system designs and assessing the design's ability to meet the system requirements. A systemic approach is required for a coherent and well-running system. Bottom-Up or Top-Down approach is required to take into account all related variables of the system. A designer uses the modelling languages to express the information and knowledge in a structure of a system that is defined by a consistent set of rules and definitions. The designs can be defined in graphical or textual modelling languages. System-level technical requirements describe the users' needs and provide information for the finished system to meet legal restrictions, adhere to regulations, and inter-operate or integrate effectively with other systems. We can say that system design ranges from discussing the system requirements to product development. System development creates or alters the system so that the processes, practices and methodologies are changed to develop the system

4.2 DFD of the Proposed Candidate System

4.2.1 Data Flow Diagram for Candidate system-1



Description of the Proposed Candidate System-1 DFD

This Data Flow Diagram (DFD) represents the proposed candidate system 1 for an online grocery e-commerce system. The flow of data has been modified and enhanced compared to the existing system. Here's a breakdown of its components:

1. Customer Browsing and Product Sorting (Processes 1.1 & 1.2):

- **Customer Browsing (process 1.1):** Customers browse the product catalog and view products.
- **Product Sorting & Flash Sales (process 1.2):** Products are sorted based on certain criteria, such as flash sales, to enhance the customer's shopping experience.

2. Order Process (Edit Order - Process 2):

Customers place their order, and the system collects Order Details and checks them through the Edit Order process. If a product is not available, the system communicates with the Market Place to procure or assemble the product.

3. Assembling Products (Process 9.1):

Products are assembled from the Warehouse or Market Place as required. Packers check the products before proceeding to shipment. The Warehouse provides information about stock availability, and the Product Availability Info is confirmed.

4. Advance Payment (Process 4):

In the new system, there's an additional step for Advance Payment. The order proceeds only after this payment verification.

5. Payment (Processes 5, 6, 7):

- **Payment (process 5):** Customers make payments for their orders.
- **SMS Verification (process 6):** After payment, the customer receives an SMS to verify the transaction.

- **Create Invoice (process 7):** Once payment is verified, an invoice is created and sent to the customer.

6. Verification of Shipment and Payment (Processes 8, 9.2, and 10):

- **Verify Payment (process 8):** Ensures the payment is successful before proceeding.
- **Verify Product List (process 9.2):** The system verifies the list of products to ensure everything is correct.
- **Verify Shipment (process 10):** Final verification of the shipment before handover to the delivery team.

7. Delivery Process (Processes 11.1, 11.2, 11.3):

- **Delivery (process 11.1):** After shipment verification, the order is handed over to the delivery team.
- **Track Delivery Performance (process 11.2):** Delivery performance, including time and driver performance, is tracked.
- **Bonus Payment (process 11.3):** If the delivery meets performance standards, the delivery personnel may receive bonus payments.

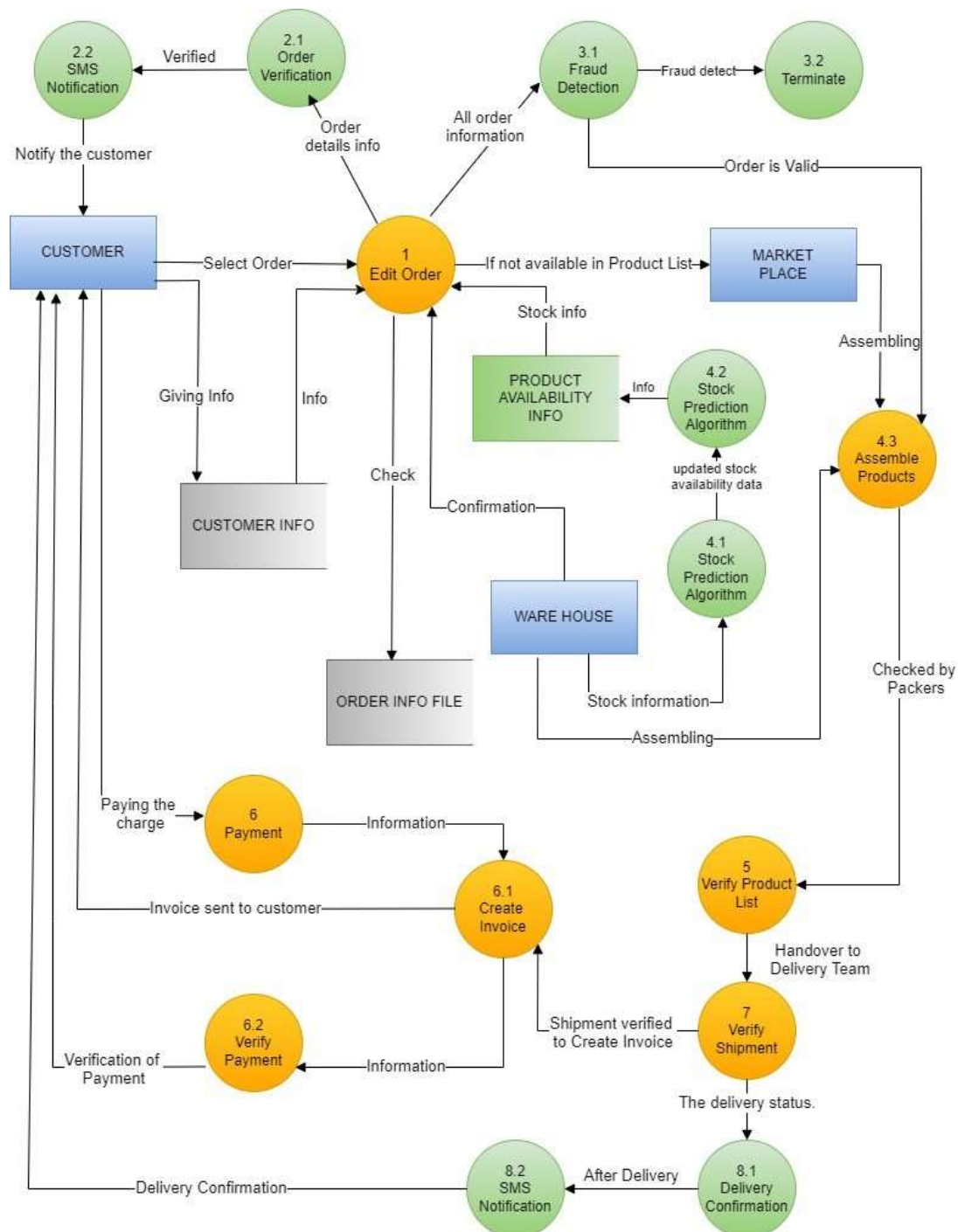
Additionally, There is an Order Confirmation SMS (process 3) that informs customers that their order has been confirmed. The Order Info File stores all order details, including customer information and payment status, updating as the order progresses. Deliveryman Info is updated once the order is delivered, providing a performance review.

Key Differences from the Existing System:

- Introduction of Customer Browsing, Product Sorting, and Flash Sales to enhance the shopping experience.
- Advance Payment step to secure order processing.
- SMS Verification as an additional layer of security after payment.
- Enhanced tracking of Delivery Performance and the possibility of Bonus Payments for delivery personnel.

In summary, the proposed candidate system adds several enhancements to improve the order management process, payment security, and delivery performance, making the system more efficient and user-friendly.

4.2.2 Data Flow Diagram for Candidate system-2



Description of the proposed DFD for candidate system-2

This Data Flow Diagram (DFD) illustrates the proposed workflow for the updated grocery e-commerce system, highlighting additional checks, fraud detection, and predictive algorithms. Here's a detailed description:

1. Edit Order (Process 1)

- Customer selects and edits their order.
- The system verifies stock availability from Product Availability Info and checks with the Warehouse for stock updates.
- If an item is not available in the Product List, the system pulls the information from an external Market Place to handle product assembly.

2. Order Verification (Process 2.1)

- Before proceeding with the order, it goes through an Order Verification process to ensure its validity.
- If verified, an SMS Notification (2.2) is sent to the customer.

3. Fraud Detection (Process 3.1) & Termination (Process 3.2)

- The system includes a Fraud Detection mechanism that checks for any potential fraudulent activity in the order.
- If fraud is detected, the process terminates at Terminate (3.2).
- If the order is deemed valid, it proceeds further.

4. Stock Prediction Algorithm (Process 4.1 & 4.2)

- A Stock Prediction Algorithm is used at two stages: one in the Warehouse and another for Product Availability Info.
- The algorithm updates the stock availability in real-time, using predictive data to manage inventory efficiently.

5. Assemble Products (Process 4.3)

- Products are then Assembled by checking with the Warehouse and Market Place.
- Packers verify the assembled products before moving forward.

6. Verify Product List (Process 5)

- Once assembled, the Product List is verified to ensure the correct items are included.
- The items are then handed over to the Delivery Team.

7. Payment (Process 6)

- The Customer proceeds with the Payment, after which an Invoice (6.1) is created.
- Payment information is also sent for verification.

8. Verify Payment (Process 6.2)

- After the payment is made, the system verifies the payment details.

9. Verify Shipment (Process 7)

- After verifying the product list, the shipment is verified and updated in the system.

10. Delivery and SMS Notifications (Processes 8.1 & 8.2)

- Once the shipment is delivered, the system sends a Delivery Confirmation (8.1) and an SMS Notification (8.2) to the customer to inform them about the delivery status.

Supporting Data Stores and Entities:

- **Customer Info:** Holds customer details, used throughout the order, verification, and payment processes.
- **Order Info File:** Central repository for all order-related information, updated at each step.

- **Product Availability Info:** Manages product stock data and runs the stock prediction algorithm.
- **Warehouse:** Handles stock updates, assembling, and verification before shipping the products.

Key Differences in the Proposed System:

- Order Verification and Fraud Detection have been added to ensure the legitimacy of orders before processing.
- Stock Prediction Algorithm helps optimize inventory management by predicting stock availability and needs.
- Enhanced notification system, with SMS alerts both at order verification and after delivery, keeping customers informed throughout.
- Streamlined processes for better integration of payment verification, shipment verification, and delivery.

This DFD reflects a more secure, automated, and customer-centric system, incorporating real-time data analysis, fraud prevention, and predictive stock management to enhance the grocery e-commerce process.

The way Proposed Candidate System Solves Key Issues in the Existing system of chaldal

The Proposed Candidate System addresses several key issues present in the existing system. Here's a breakdown of the main issues and how the new system resolves them:

1. Lack of Browsing and Sorting Features

- **Issue in Existing System:** The current system does not provide advanced browsing or product sorting options for customers, which limits their ability to quickly find products based on preferences such as sales or product type.
- **Solution in Proposed System:**

- **Customer Browsing (1.1):** The new system introduces a browsing feature, allowing customers to view available products more easily.
- **Product Sorting & Flash Sales (1.2):** Products are sorted based on criteria like promotions or flash sales, enhancing the shopping experience by making it easier to find deals.

2. No Advance Payment Option

- **Issue in Existing System:** The existing system only verifies payment after products are verified and assembled, leading to potential delays or issues if payment verification fails later in the process.
- **Solution in Proposed System:**
 - **Advance Payment (4):** The new system introduces an advance payment step, ensuring payment is received before the order is processed further. This prevents unnecessary processing of unpaid orders and avoids delays.

3. No SMS Verification for Payment

- **Issue in Existing System:** Payment verification happens without additional customer confirmation, which could lead to issues like unauthorized payments or failed notifications.
- **Solution in Proposed System:**
 - **SMS Verification (6):** After payment is made, customers receive an SMS to confirm the payment, ensuring an extra layer of security and providing a clear communication channel to the customer about their transaction status.

4. Unclear Delivery Performance Tracking

- **Issue in Existing System:** The delivery performance, such as delivery speed or driver reliability, is not tracked, leading to inconsistent service levels and lack of incentives for delivery personnel.
- **Solution in Proposed System:**

- **Track Delivery Performance (11.2):** The new system tracks delivery metrics like delivery time and driver performance, allowing for better monitoring and improvement of service levels.
- **Bonus Payment (11.3):** To incentivize better performance, the system introduces a bonus payment feature for delivery personnel based on performance, encouraging timely and efficient deliveries.

5. Lack of Real-Time Product Verification

- **Issue in Existing System:** The product verification process is not streamlined, and any mismatch between the order and available products is handled after product assembly, leading to inefficiencies.
- **Solution in Proposed System:**
 - **Verify Product List (9.2):** In the proposed system, the product list is verified before proceeding to shipping, ensuring that the correct items are being shipped to the customer. This reduces errors and improves accuracy.

6. Inefficiencies in Order and Shipment Handling

- **Issue in Existing System:** The existing system may not effectively handle large volumes of orders, leading to inefficiencies in order processing, shipment preparation, and fulfillment.
- **Solution in Proposed System:**
 - **Streamlined Processes:** By introducing a more organized order flow, including the verification of orders, advance payment, and assembling, the proposed system reduces bottlenecks and ensures a smoother flow of data and goods.
 - **Track Shipment and Delivery (Processes 10 and 11.1):** The tracking of shipments and real-time handover to delivery teams are better managed, resulting in faster fulfillment and fewer errors in deliveries.

7. Lack of Market Integration for Out-of-Stock Products

- **Issue in Existing System:** When products are not available in stock, the existing system relies on manual intervention to source products from other markets, causing delays.
- **Solution in Proposed System:**
 - **Market Place Integration:** The proposed system has a well-defined integration with the Market Place for assembling and sourcing products when they are unavailable, reducing the time required to fulfill orders and improving product availability.

8. Inconsistent Communication of Order Status

- **Issue in Existing System:** Customers might not receive real-time updates on their order status, leading to confusion and dissatisfaction.
- **Solution in Proposed System:**
 - **Order Confirmation SMS (3):** The system now automatically sends an order confirmation SMS once the order is placed, keeping customers informed immediately.
 - **Invoice and Payment Updates (Processes 7 & 6):** Invoice generation and payment updates are communicated to the customer in a more structured way, ensuring transparency throughout the process.

9. No Performance-Based Delivery Incentives

- **Issue in Existing System:** The current system does not incentivize delivery personnel based on their performance, potentially leading to inconsistent service quality.
- **Solution in Proposed System:**
 - **Bonus Payment for Delivery (11.3):** The new system introduces performance-based bonuses for delivery personnel, motivating them to provide better service and complete deliveries on time.

Summary of Improvements: The Proposed Candidate System 1 addresses key issues by:

- Introducing features like browsing, sorting, and flash sales to enhance the customer experience.
- Implementing an Advance Payment system to avoid unpaid orders progressing through the workflow.
- Adding SMS verification and more robust delivery tracking to improve transparency and performance.
- Integrating the Market Place for out-of-stock products, ensuring faster product sourcing.
- Offering incentives for delivery personnel based on performance, improving delivery times and customer satisfaction.

Overall, the proposed system streamlines processes, improves communication, and enhances customer experience, leading to a more efficient and reliable e-commerce system.

4.3 Database Design

A well-structured organization relies on an efficient database system to streamline its operations and manage data effectively. As the largest e-commerce grocery delivery service in Bangladesh, Chaldal Ltd. necessitates a comprehensive database to support its operation and maintenance system. This database is crucial for facilitating seamless data flow from product storage to delivery, providing valuable insights that enhance operational processes.

4.3.1 Objectives

The objectives of our database design are essential for ensuring its functionality, reliability, and contribution to the organization's success:

1. **Efficient Data Storage:** Our database will enable the mass storage of relevant data, allowing for the efficient management of information related to inventory, orders, and customer interactions. This ensures that all necessary data is readily available for operational needs and strategic decision-making.

2. **User Accessibility:** By providing authorized users with easy access to data, we aim to enhance productivity. Quick retrieval of information will empower staff to respond effectively to customer inquiries, manage inventory, and streamline operations.
3. **Data Integrity and Security:** Maintaining high data integrity is paramount. The database will implement strict validation rules and checks to ensure accuracy and consistency in data entries. Additionally, robust security measures will protect sensitive information from unauthorized access and potential threats, safeguarding the company's reputation.
4. **Reduction of Redundancy:** The design will focus on eliminating duplicated data entries, which can lead to inconsistencies and inefficiencies. By centralizing data storage and implementing normalization techniques, we will enhance the overall efficiency and accuracy of the database.
5. **Multi-User Support:** Our database will be designed to support simultaneous access for multiple users, ensuring that team members can collaborate effectively without performance degradation. This will facilitate a dynamic work environment where operations can run smoothly.
6. **Scalability:** The database will be built with scalability in mind, ensuring that it can grow and adapt to the increasing demands of Chaldal Ltd. as it expands. This includes the capability to accommodate additional data types, users, and functionalities as needed.
7. **Comprehensive Reporting:** We will integrate interrelated data to generate insightful reports that provide valuable analysis of operational performance. These reports will assist management in making informed decisions, identifying trends, and optimizing processes to improve customer satisfaction.

By focusing on these objectives, our database design will significantly enhance the operational efficiency of Chaldal Ltd. It will not only support current processes but also position the company for future growth and improved customer service, reinforcing its status as a leader in the e-commerce grocery sector.

4.3.2 Entity Relationship Diagram

The ER diagram or (Entity Relational Diagram) is the diagram of a top level conceptual data model. E-R modelling will help greatly to analyse data requirements graphically to produce an easy and more maintainable database. An ER diagram shows the relationship among entity sets. An entity set is a group of similar entities and these entities can have attributes. In terms of DBMS, an entity is a table or attribute of a table in a database, so by showing relationship among tables and their attributes, ER diagram shows the complete logical structure of a database. The E-R diagram basically represents between the tables.

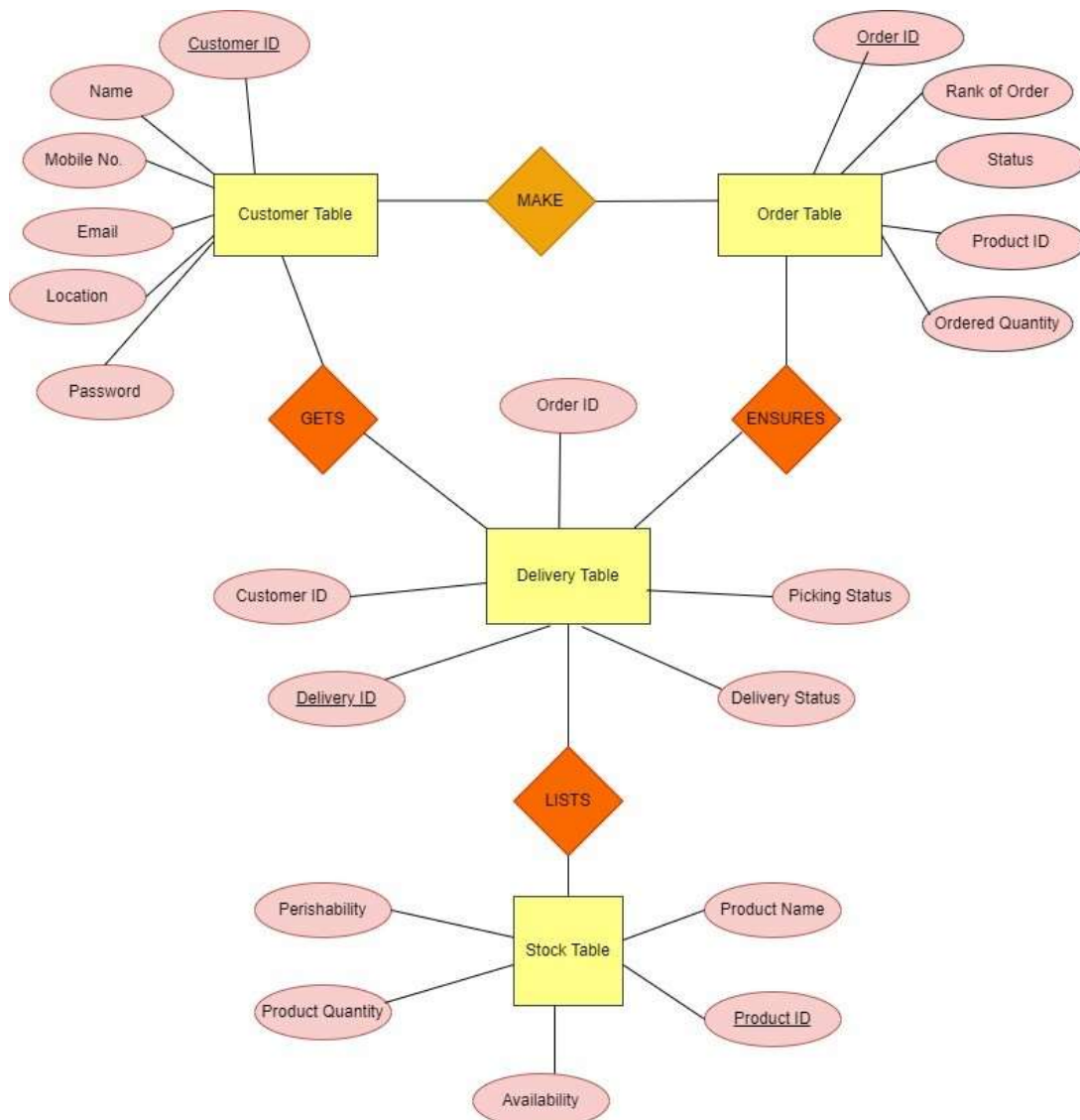


Figure 4.3.2 : Structure chart of Delivery management

4.3.3 Data type and size according to the ER diagram

4.3.3.1 Customer Table

In Table 4.3.1, the information of a customer and the information regarding customer's order are tabulated. Here, in total 7 attributes are used. Among which 'Customer ID' is the primary key and 'Order ID' is the foreign key referenced to Order Table. The attributes of this table are –

Customer ID (Primary Key) [Type: varchar Size: 10]

Name [Type: varchar Size: 30]

Mobile No. [Type: varchar Size: 11]

Email [Type: varchar Size: 30]

Password [Type: varchar Size: 8]

Location [Type: point]

Order ID (Foreign Key) [Type: varchar Size: 10]

4.3.3.2 Stock Table

In Table 4.3.2, the information of the inventory and the information regarding product's information are tabulated. Here, in total 6 attributes are used. Among which 'Product ID' is the primary key. The attributes of this table are-

Product ID (Primary Key) [Type: varchar Size: 10]

Product Name [Type: varchar Size: 30]

Product Quantity [Type: int Size: 11]

Perishability [Type: varchar Size: 5]

Availability [Type: varchar Size: 5]

4.3.3.3 Order Table

In Table 4.3.3 table is to track all the order information. Here, in total there are 5 attributes where Order ID is the primary key and Product ID is the foreign key referenced to the Stock table. The attributes are -

Order ID (Primary Key) [Type: varchar Size: 10]

Status [Type: varchar Size: 10]

Rank of Order [Type: int Size: 11]

Product ID (Foreign Key) [Type: varchar Size: 10]

Ordered Quantity [Type: int Size: 11]

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 CustomerID	varchar(10)	utf8mb4_general_ci		No	None			Change Drop More
☐	2 Name	varchar(30)	utf8mb4_general_ci		Yes	NULL			Change Drop More
☐	3 MobileNo	varchar(11)	utf8mb4_general_ci		Yes	NULL			Change Drop More
☐	4 Email	varchar(30)	utf8mb4_general_ci		Yes	NULL			Change Drop More
☐	5 Password	varchar(8)	utf8mb4_general_ci		Yes	NULL			Change Drop More
☐	6 Location	point			Yes	NULL			Change Drop More
☐	7 OrderID	varchar(10)	utf8mb4_general_ci		Yes	NULL			Change Drop More

Table 4.3.1 : Customer Table

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 ProductID	varchar(10)	utf8mb4_general_ci		No	None			Change Drop More
☐	2 ProductName	varchar(30)	utf8mb4_general_ci		Yes	NULL			Change Drop More
☐	3 ProductQuantity	int(11)			Yes	NULL			Change Drop More
☐	4 Perishability	varchar(5)	utf8mb4_general_ci		Yes	NULL			Change Drop More
☐	5 Availability	varchar(5)	utf8mb4_general_ci		Yes	NULL			Change Drop More

Table 4.3.2 : Stock Table

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 OrderID	varchar(10)	utf8mb4_general_ci		No	None			Change Drop More
☐	2 Status	varchar(10)	utf8mb4_general_ci		Yes	NULL			Change Drop More
☐	3 RankOfOrder	int(11)			Yes	NULL			Change Drop More
☐	4 ProductID	varchar(10)	utf8mb4_general_ci		Yes	NULL			Change Drop More

Table 4.3.3 : Order Table

4.3.3.4 Delivery Table

Here the table reflects the system and sections included regarding the Delivery System. Here in total, there are 5 attributes. 'Customer ID' and 'Order ID' are the foreign keys here referenced to customer table and order table respectively. The attributes are –

Delivery ID (Primary Key) [Type: varchar Size: 10]

Delivery Status [Type: varchar Size: 10]

Picking Status [Type: varchar Size: 10]

Order ID (Foreign Key) [Type: varchar Size: 10]

Customer ID (Foreign Key) [Type: varchar Size: 10]

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐ 1	DeliveryID 	varchar(10)	utf8mb4_general_ci		No	None			 Change  Drop  More
☐ 2	DeliveryStatus	varchar(10)	utf8mb4_general_ci		Yes	NULL			 Change  Drop  More
☐ 3	PickingStatus	varchar(10)	utf8mb4_general_ci		Yes	NULL			 Change  Drop  More
☐ 4	OrderID 	varchar(10)	utf8mb4_general_ci		Yes	NULL			 Change  Drop  More
☐ 5	CustomerID 	varchar(10)	utf8mb4_general_ci		Yes	NULL			 Change  Drop  More

Figure 4.3.4 : Delivery Table

4.4 Form Design

Form design is a key element of system analysis. It focuses on creating user interfaces that simplify data entry, display, and manipulation within a software system. A well-crafted form can improve user experience, boost data accuracy, and optimize business processes. Since the proposed candidate system is focused on ordering, assembly, and delivery, the current forms do not need to be modified.

Login/Signup:

Chaldal.com's login and registration form is designed to provide a seamless and user-friendly experience. Users can quickly sign up using basic details like their email, phone number, and password. The registration process includes SMS verification to ensure security and prevent fraudulent activity. For returning users, the login process is simple, requiring only their registered email or phone number and password. The system also offers a "Forgot Password" feature to help users easily recover their accounts.

Personal Information and Address:

The personal information form includes a field for entering the user's phone number, with an option to mark it as the "Default Phone." Users can provide their complete name

in the designated field. Additionally, there is space for entering the user’s residential address, which includes the street name and any additional details. A drop-down menu allows users to select their city to ensure accurate delivery. Gender selection options are available, including Male, Female, and Others. Finally, the date of birth is captured through three fields—Day, Month, and Year—ensuring accurate entry of the user’s birth date.

← Mobile Login

Enter your mobile number

You will receive a one time pin as an SMS

+880 Mobile Number

Login with OTP

Other Login Options

← Login

Login to Chaldal

Your email address

Your password

Login to Chaldal

Forgot your password?

OR

Login with Facebook

Figure 4.4.1 : Login and Sign Up form

Your Profile

Name

01538052523

Verify your email address & Get 1 free delivery

Email Address

VERIFY

Phone Number

+8801538052523

Gender

--Select Gender--

Date of Birth (YYYY-MM-DD)

Address Book

+ New Address

Figure 4.4.2 : Personal Information and Address

Order Form

The order form consists of various fields for capturing essential details about the user's purchase. First, there is a section for entering the product name, allowing users to specify the item they wish to order. Users can select the quantity of the desired product in a designated field, ensuring accurate order processing. The form also includes options for selecting the preferred delivery method, such as standard or express shipping. Additionally, users can provide any special instructions for their order in a comments section. Finally, a summary of the order displays the total price, including applicable taxes and fees, ensuring transparency before finalizing the purchase.

The screenshot displays a mobile application's order form. It features three main sections: 'Select a Delivery Address' with a location pin icon and a '+ New Address' button; 'Preferred Delivery Time' with a clock icon and a message stating 'Slot selection will be available after selecting an address'; and 'Add reusable bags?' with a bag icon, an information icon, and a toggle switch. At the bottom, it shows 'Payment options available on the next page', a 'Proceed' button, and a total price of '₹ 6,086' with a note '₹0 Delivery charge included'. A small disclaimer at the bottom reads 'By clicking/tapping proceed, I agree to Chalda's Terms of Services'.

Figure 4.2.3 : Order Form

Payment Method

The payment method section provides users with options to select their preferred method of payment for their order. Users can choose from various options, including credit or debit card payments, cash on delivery, or other available payment methods. A straightforward interface allows users to select their preferred option with ease, ensuring a seamless checkout experience. Clear instructions are provided for each payment method to facilitate smooth transactions and enhance user convenience.

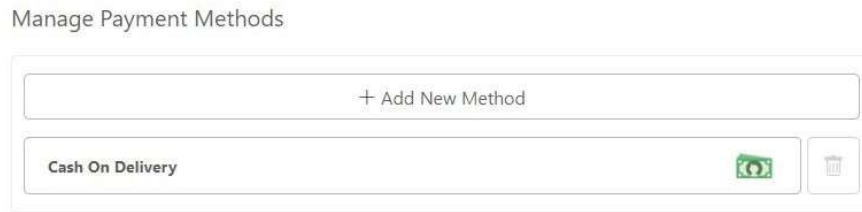


Figure 4.4.4: Payment Process

4.5 Conclusion

The system design phase plays a pivotal role in the software development life cycle (SDLC), where the abstract concepts and requirements from the analysis phase are transformed into a concrete blueprint for implementation. This phase is essential for shaping the foundation of the software system by defining both the overarching architecture and the detailed specifications that guide developers throughout the implementation process.

System design involves an intricate process of translating user and business requirements into a structured design. This design encompasses various components, including the architecture that outlines the system's major modules, their functions, and their interconnections. The goal is to ensure that each part of the system can seamlessly interact and work together, maintaining a coherent structure while optimizing performance. Design decisions made during this phase determine key aspects such as data structures, algorithms, and how information will flow within the system.

Throughout this phase, design principles such as modularity, cohesion, and encapsulation are applied to ensure that the system is robust, flexible, and easy to maintain. Additionally, adherence to industry standards and best practices guarantees that the system is reliable and capable of meeting both current and future demands. In conclusion, the system design phase is an essential step in creating a well-rounded software solution, ensuring that all components work together efficiently while aligning with business goals and user expectations.